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Month, Year

To my family.

Acknowledgments

I would like to thank my supervisor, **Prof. Supervisor Name**, for the guidance and support throughout this work.

Resumo

Este trabalho aborda o problema de …

Palavras-chave: palavra-chave 1, palavra-chave 2, palavra-chave 3.

Abstract

This thesis addresses the problem of …

Keywords: keyword 1, keyword 2, keyword 3.

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List of Acronyms

Acronym	Definition
ACDC	Automated Cardiac Diagnosis Challenge
AHA	American Heart Association
ASSD	Average Symmetric Surface Distance
DSC	Dice Similarity Coefficient
DCM	Dilated Cardiomyopathy
ED	End-Diastole
EDT	Euclidean Distance Transform
ES	End-Systole
GNN	Graph Neural Network
HCM	Hypertrophic Cardiomyopathy
HD95	95th-percentile Hausdorff Distance
ICC	Intraclass Correlation Coefficient
INR	Implicit Neural Representation
IoU	Intersection over Union
LV	Left Ventricle
LoA	Limits of Agreement
MAE	Mean Absolute Error
MRI	Magnetic Resonance Imaging
NOR	Normal cohort
PCA	Principal Component Analysis
PDE	Partial Differential Equation
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RMSE	Root Mean Squared Error
RV	Right Ventricle
SAX	Short-Axis
SDF	Signed Distance Function
SSM	Statistical Shape Model

CHAPTER 1

Introduction

This chapter introduces the clinical and technical motivation for reconstructing three-dimensional left-ventricular geometry from sparse cardiac MRI. It defines the problem, research questions, objectives, and contributions of the thesis.

1.1. Motivation

The heart keeps blood moving through the body. The left ventricle (LV) has a central role because it pumps oxygen-rich blood into the aorta and from there to the rest of the body. Its size, shape, motion, and wall thickness can therefore provide important information about cardiac function. Changes in LV shape and wall thickness can also be associated with disease (de Marvao et al., 2014; Grossman et al., 1974).

Cardiac magnetic resonance imaging (MRI) allows the LV to be observed without an invasive procedure. Short-axis (SAX) images show the heart as a stack of slices from the base to the apex. These images make it possible to identify the inner and outer boundaries of the myocardium, which is the muscular wall of the heart. However, the slices alone do not provide a complete three-dimensional view. A 3D model can describe the full LV shape and support measurements in the regions between slices. Consistent descriptions of LV anatomy are useful in large imaging studies and automated analysis (Bai et al., 2015, 2018).

Myocardial wall thickness is one such measurement. A wall that is unusually thick, thin, or uneven can contain useful information about the state of the heart. In clinical images, wall thickness is often measured only on the acquired slices. A smooth 3D representation could provide a more complete and more easily visualised measurement across the LV. This thesis studies how sparse SAX observations can be used to reconstruct the LV in three dimensions and to measure myocardial wall thickness.

1.2. Problem Statement

SAX cardiac MRI is useful, but it is not a detailed 3D scan of the heart. The slices are separated along the long axis of the LV, leaving part of the shape unobserved between them. This makes 3D reconstruction difficult. Simply connecting the slices can create rough surfaces, gaps, or a myocardial wall that does not represent the anatomy well.

The problem has two linked parts. First, the unobserved geometry between slices must be completed without removing patient-specific shape. Second, two related surfaces must be reconstructed. The endocardium is the inner boundary of the LV cavity, and the epicardium is the outer boundary of the myocardium. The space between them must remain a valid wall everywhere. Public cardiac MRI datasets usually provide images and segmentation masks for each slice, rather than complete 3D LV meshes. Such meshes are needed to train and evaluate a surface-reconstruction model directly.

This work addresses the problem with a phase-conditioned implicit neural representation (INR) model trained using both synthetic LV shapes and meshes derived from real cardiac MRI segmentations. The model receives sparse contours as input and predicts continuous 3D LV surfaces. The outer surface is constrained to remain outside the inner surface, producing a myocardial wall with positive thickness. The reconstructed geometry is then used to study wall-thickness measurements and compare them with measurements from segmentation-derived geometry.

1.3. Research Questions

This work addresses three research questions. **RQ1** asks whether a model can reconstruct smooth, closed 3D LV surfaces from sparse SAX contours. **RQ2** asks whether the model can represent the endocardium and epicardium while maintaining a valid myocardial wall at end-diastole (ED) and end-systole (ES). **RQ3** asks how closely wall-thickness measurements from the reconstructed LV agree with measurements from segmentation-derived geometry.

1.4. Objectives

The first objective is to prepare synthetic and real cardiac MRI data in a shared contour-based format. The second objective is to develop a phase-conditioned signed distance function (SDF) model that reconstructs the inner and outer LV surfaces from sparse SAX contours. A further objective is to encourage smooth, closed reconstructed surfaces and prevent collapse of the endocardial and epicardial implicit fields. Finally, the reconstructed surfaces are evaluated against segmentation-derived meshes, and several methods for measuring local and regional wall thickness are compared.

1.5. Contributions

This thesis provides a data pipeline that converts synthetic LV shapes and real cardiac MRI segmentations into a shared representation of sparse SAX contours and 3D meshes. It also presents a phase-conditioned signed-distance model that predicts the endocardial and epicardial LV surfaces while preserving a positive wall between them. Finally, it evaluates the reconstructed geometry on data not used for training and compares local and regional wall-thickness methods on reconstructed and segmentation-derived LV geometry.

1.6. Document Structure

Chapter 2 reviews cardiac MRI reconstruction, statistical shape models, implicit representations, and wall-thickness measurement. Chapter 3 describes the data preparation, model, training process, and evaluation protocol. Chapter 4 reports the reconstruction and wall-thickness results and discusses their validity. Finally, Chapter 5 summarises the answers to the research questions and the main directions for future work.

CHAPTER 2

Literature Review

This chapter presents the review methodology used to identify relevant literature and synthesises the related work on three-dimensional LV reconstruction from cardiac MRI and myocardial wall-thickness measurement.

2.1. Review Methodology

The review used a structured search to identify work that informs four parts of the thesis: sparse cardiac MRI, 3D shape reconstruction, geometric learning, and myocardial wall-thickness measurement. The purpose was to establish the design choices and research gap, rather than to provide an exhaustive systematic review.

2.1.1. Search strategy

The search was conducted across three major bibliographic databases: Scopus, PubMed, and IEEE Xplore, supplemented by Google Scholar for citation chaining. The review includes foundational publications from 2000 onwards and selected studies available up to 2026. It prioritises work from the last decade while retaining methods essential for theoretical context. The review is selective and does not claim exhaustive coverage of the literature published through 2026.

Search terms were grouped into four thematic categories and combined using Boolean operators. The first category covered anatomy and imaging terms ("cardiac MRI", "left ventricle", "short-axis", "SAX", "myocardium"). The second addressed reconstruction ("3D reconstruction", "mesh reconstruction", "shape model", "implicit surface", "signed distance function"). The third targeted learning methods ("deep learning", "graph neural network", "point cloud", "neural representation"). The fourth focused on measurement ("wall thickness", "myocardial thickness", "Laplace equation", "transmural").

The refined query applied consistently across databases was:
("cardiac MRI" OR "left ventricle" OR "short-axis") AND ("3D reconstruction" OR "statistical shape model" OR "signed distance function" OR "graph neural network" OR "implicit representation") AND ("wall thickness" OR "mesh" OR "deep learning")

2.1.2. Inclusion and exclusion criteria

Studies were included if they: (i) addressed three-dimensional reconstruction or shape modelling of the LV, (ii) proposed or evaluated methods for myocardial wall-thickness measurement, (iii) developed geometric deep learning techniques applicable to anatomical meshes, or (iv) introduced benchmark cardiac MRI datasets used for LV analysis.

Studies were excluded if they: (i) focused exclusively on segmentation without reconstruction or shape analysis, (ii) addressed organs other than the heart without methodological generalisability, (iii)

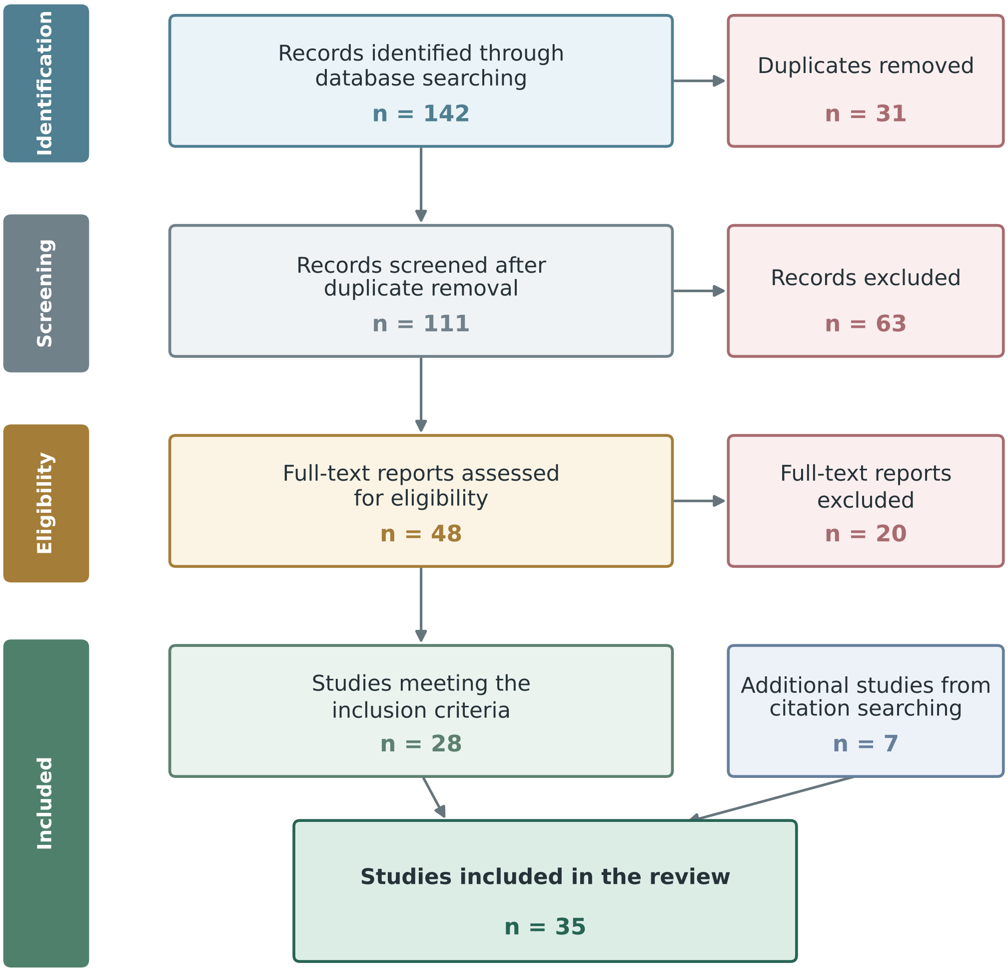

Figure 2.1. PRISMA flow diagram summarising the selection of the 35 publications included in the review.

lacked peer review or sufficient methodological description, or (iv) were published before 2000 unless they constituted foundational work directly relevant to the methods reviewed.

2.1.3. Results

The search returned 142 records. After duplicate removal and screening, 28 studies met the selection criteria; backward citation searching added 7, giving a final corpus of 35 publications. Figure 2.1 summarises this process.

The selected studies cover cardiac MRI data and segmentation, statistical shape models, learned 3D reconstruction, and wall-thickness measurement. The next section relates these areas to the specific problem of recovering paired LV surfaces from sparse contours.

2.2. Technical Background

This section introduces the geometric concepts needed to compare the reviewed methods. It separates what is observed in cardiac MRI from what a reconstruction method must infer, and then contrasts the main ways in which the resulting 3D anatomy can be represented.

2.2.1. From SAX slices to paired LV surfaces

Cine cardiac MRI records the heart at several points in the cardiac cycle. End-diastole (ED) corresponds approximately to the phase at which the LV cavity is largest, while end-systole (ES) corresponds to its smallest state. These two phases provide a compact description of ventricular contraction and are the phases labelled in commonly used benchmarks such as ACDC (Bernard et al., 2018).

In a short-axis acquisition, each image plane cuts across the ventricular long axis. The in-plane anatomy is sampled in detail, but consecutive planes are separated by a larger distance. The resulting stack is therefore anisotropic: spatial resolution depends on direction (Petitjean & Dacher, 2011). A segmentation can identify the visible myocardium on every acquired plane, but it does not directly observe how the boundary bends, tapers, or changes between those planes.

Two contours are needed on each slice. The endocardial contour bounds the blood cavity, while the epicardial contour bounds the outer myocardium. In three dimensions these contours become two nested surfaces. Their relationship is as important as either surface alone: an intersection, reversal, or local collapse would imply an invalid myocardial wall. Reconstructing the surfaces separately can therefore create a plausible cavity and outer boundary while still failing to produce a valid wall between them.

The reconstruction problem is underdetermined because many smooth surfaces can pass through the same contour rings. A method must decide how much shape is inferred from the observations and how much is supplied by a prior learned from other hearts. Strong smoothing gives stable surfaces but may remove real regional variation. A strong population prior can fill large gaps but may pull an abnormal ventricle towards common anatomy. Conversely, fitting every contour closely can preserve the observations while reproducing segmentation noise or creating abrupt changes between slices.

For this reason, reconstruction quality has several dimensions. The surface should remain close to the observed anatomy, form a closed geometry where a volume is required, preserve patient-specific differences, and maintain a valid wall at both cardiac phases. No single overlap or surface-distance metric tests all of these properties. The evaluation must combine geometric distance, volumetric agreement, topology, and the stability of measurements derived from the wall.

2.2.2. Representations of three-dimensional anatomy

Three-dimensional cardiac anatomy can be stored as a voxel volume, an explicit surface mesh, or an implicit field. A voxel volume follows the image grid and is convenient for segmentation, overlap measures, and distance transforms. Its spatial detail is limited by the grid resolution, however, and interpolation is still needed where the acquired slices are far apart.

A triangular mesh represents the boundary directly through vertices, edges, and faces. It supports surface distances, enclosed-volume calculations, regional partitioning, and colour maps of local measurements. Meshes with consistent topology and vertex correspondence are also useful for population statistics. Their quality depends on how the vertices are connected and distributed, and operations such as smoothing, remeshing, or hole repair can alter the geometry being evaluated.

Statistical shape models provide one way to generate corresponded meshes. They describe each anatomy as a mean shape plus weighted population modes (Cootes et al., 1995; Heimann & Meinzer, 2009). This gives a compact and controllable shape space, but it links the output to the topology and variation present in the atlas. An SSM is therefore well suited to supplying a geometric prior or synthetic training examples, but it is not an independent observation of a new patient's anatomy.

An implicit representation defines geometry through a function evaluated at continuous coordinates. An occupancy field predicts whether a point is inside or outside the anatomy, whereas a signed distance function also represents the distance to the boundary (Mescheder et al., 2019; Park et al., 2019). The surface is the zero-level set of this function and can be extracted on a grid with an algorithm such as Marching Cubes (Lorensen & Cline, 1987). Because the function is continuous, the extraction resolution is not fixed by an output template.

For a true signed distance function, the gradient magnitude is one almost everywhere away from singular points. Eikonal regularisation encourages a neural field to retain this property even when dense distance targets are unavailable (Gropp et al., 2020). Fourier coordinate features can improve the recovery of finer spatial variation that a standard multilayer perceptron tends to smooth (Tancik et al., 2020). These properties make implicit fields attractive for interpolating sparse contours, although the final mesh still depends on field quality, extraction resolution, and post-processing.

When two nested surfaces are needed, two independent implicit functions do not guarantee their ordering. A coupled representation can instead define one surface relative to the other through a positive separation. This changes the problem from predicting two unrelated boundaries to predicting a reference surface and a constrained wall. The reviewed literature provides the individual elements of this approach, but their integration for phase-conditioned LV reconstruction is the specific gap examined later in this chapter.

2.3. Related Work

This section synthesises the reviewed literature, organised by thematic area.

2.3.1. Cardiac MRI segmentation and datasets

Short-axis (SAX) cine MRI is the standard protocol for left-ventricle (LV) evaluation, producing 2D slices perpendicular to the long axis with high in-plane resolution (1–2 mm) but large inter-slice gaps (6–10 mm) (Cerqueira et al., 2002; Petitjean & Dacher, 2011). This anisotropic acquisition creates the central reconstruction problem: the observed contours are detailed within each slice, but the geometry between slices, especially near the apex and base, must be inferred. The AHA 17-segment model provides a common anatomical frame for reporting the resulting regional measurements (Cerqueira et al., 2002).

Modern segmentation networks can identify endocardial and epicardial boundaries reliably on the acquired slices. U-Net established the encoder–decoder pattern used by many medical segmentation systems, while nnU-Net showed that much of their configuration can be adapted automatically to a dataset (Isensee et al., 2021; Ronneberger et al., 2015). These methods improve the quality of the observed contours, but they do not by themselves determine the unobserved 3D surface between slices.

The distinction between segmentation and reconstruction is important here. U-Net combines broad image context with fine spatial detail through skip connections, making it suitable for delineating thin myocardial boundaries. The self-configuring design of nnU-Net reduces the need to tune preprocessing, network scale, and post-processing separately for each dataset. Both approaches still optimise agreement on sampled image planes. A high slice-wise overlap can therefore coexist with an uncertain surface between planes, because no image evidence directly constrains that part of the anatomy.

The ACDC benchmark provides expert ED and ES labels across several clinical groups, including normal and hypertrophic ventricles (Bernard et al., 2018). The M&Ms challenge extends this setting across centres, scanner vendors, and diseases, showing that performance on one source does not guarantee reliable transfer to another (Campello et al., 2021). Together, these datasets support model development on varied clinical images while also exposing the limits of generalisation.

ACDC contains 100 cases divided among five diagnostic groups and supplies labels at the two phases with the largest functional contrast, ED and ES. It is therefore useful for studying whether a reconstruction changes appropriately with cardiac phase and whether abnormal shape is retained. M&Ms contains 375 studies from several centres and scanner vendors. Its principal lesson is that variation in acquisition can be as important as variation in anatomy: a method that performs well on one source may lose accuracy on unseen data. This limits claims based on any single evaluation cohort and supports the use of varied clinical data during fine-tuning.

At population scale, the UK Biobank provides a large and standardised cardiac MRI resource (Sudlow et al., 2015). Automated segmentation of this cohort has enabled large anatomical studies and statistical atlases (Bai et al., 2018). These resources establish two complementary forms of evidence used in this thesis: slice-wise clinical observations and population-level knowledge of plausible 3D cardiac shape.

The two forms are not interchangeable. Clinical labels retain subject-specific and pathological variation, but their through-plane geometry is only indirectly observed. A population atlas supplies complete and consistent surfaces, but it reflects the cohort and processing choices from which it was built. Combining them can reduce the weakness of either source alone, provided that the atlas is used as a prior rather than treated as patient-specific ground truth.

2.3.2. Statistical shape models

Statistical Shape Models (SSMs) encode anatomical variability by aligning a training population to point-to-point correspondence and applying PCA to extract the dominant modes of variation (Cootes et al., 1995; Heimann & Meinzer, 2009). They provide a compact description of plausible shape, but their quality depends on anatomical correspondence, alignment, and the diversity of the population used to construct them.

In a point-distribution model, the corresponding coordinates of each training shape are concatenated into one vector. Principal component analysis separates the mean anatomy from directions of population variation, and a new shape is formed by weighting those directions (Cootes et al., 1995). Restricting the weights to the range represented in the training population provides a practical plausibility constraint.

This compact representation is useful when complete meshes are scarce, although it assumes that the main variation can be described within a linear shape space.

Cardiac SSMs have progressed from multi-object ventricular models to large population atlases. Dense correspondence allows coupled structures to be represented consistently across subjects (Frangi et al., 2002), while the UK Digital Heart Project atlas captures population variation in ventricular shape and motion (Bai et al., 2015). This makes an SSM more than an average heart: its modes describe coordinated changes in size, proportion, apical form, and regional curvature.

The move from single structures to coupled cardiac models is relevant because the geometry of one surface is not independent of its neighbours. Frangi et al. (2002) used non-rigid registration to establish dense correspondence in a multi-object model, allowing related ventricular structures to vary together. Bai et al. (2015) then constructed a large-scale atlas from more than one thousand cardiac MRI studies. The first 50 modes represented about 90% of the observed shape variance, showing that a moderate latent space can retain substantial population variation while keeping all generated meshes in correspondence.

Those modes can support both population analysis and disease studies. Previous work has related cardiac shape variation to hypertrophic remodelling, normal anatomical diversity, and myocardial infarction (de Marvao et al., 2014; Medrano-Gracia et al., 2014; Suinesiaputra et al., 2018). The important point for reconstruction is that a shape prior can preserve structured variation that a simple smooth template would remove.

These studies also show why global volume alone is insufficient. de Marvao et al. (2014) found shape patterns associated with hypertrophic remodelling that were not captured fully by conventional volumetric indices. Medrano-Gracia et al. (2014) identified modes related to size, eccentricity, apical form, and regional curvature in a large multi-ethnic cohort. SSM-derived features have also been used to distinguish infarcted from healthy ventricles (Suinesiaputra et al., 2018). A useful reconstruction prior must therefore retain regional variation rather than force every case towards one mean geometry.

SSMs therefore have two useful roles. They can constrain reconstruction to a learned shape space, or they can generate synthetic geometries by sampling that space (Heimann & Meinzer, 2009; Khalafvand et al., 2018). This thesis uses the second role for pre-training: the SSM supplies complete, plausible meshes before the model is fine-tuned on sparse contours from clinical images.

Synthetic sampling increases the amount of complete geometric supervision, but it cannot introduce anatomy absent from the source atlas. It may also under-represent rare or severe pathology when the retained PCA modes favour common variation. The real-data fine-tuning stage is therefore essential: it allows the reconstruction to move beyond the synthetic prior while retaining the surface continuity learned from complete meshes. This division of roles is the main reason for combining SSM and clinical data rather than choosing one of them as the sole training source.

2.3.3. Deep learning models for 3D shape reconstruction

Classical reconstruction methods interpolate between segmented SAX slices using splines or deformable surfaces. Their result depends on the sampled contours and the assumptions imposed between slices, particularly near the apex and base (Heimann & Meinzer, 2009; Petitjean & Dacher, 2011). Simple interpolation can preserve the observations but may introduce staircase artefacts, whereas restrictive parametric surfaces can smooth away individual shape. Learned reconstruction methods address this trade-off through two main families: template deformation and implicit field prediction.

Template methods learn to displace the vertices of a fixed mesh. Pixel2Mesh and Pose2Mesh demonstrate that graph-based deformation can recover dense geometry from sparse image or landmark evidence (Choi et al., 2020; N. Wang et al., 2018). In cardiac applications, fixed correspondence is valuable for population analysis and regional measurement (Bai et al., 2015; Beetz et al., 2022; Suinesiaputra et al., 2014). The trade-off is that the output remains tied to the topology and resolution of the chosen template.

Implicit Neural Representations (INRs) take a different approach. Rather than deforming a template, they encode geometry as a continuous function whose zero-level set defines the surface. DeepSDF represents a shape class through a latent-conditioned signed distance function, while Occupancy Networks learn a continuous inside–outside decision (Mescheder et al., 2019; Park et al., 2019). Both separate the learned geometry from a fixed output mesh and allow a surface to be extracted at a chosen resolution, commonly with Marching Cubes (Lorensen & Cline, 1987).

Recent cardiac work shows why implicit representations are useful when MRI slices are far apart. In a SAX scan, the detailed shape of the ventricle is observed within each slice, but less information is available between slices. Brás et al. (2026) addressed this anisotropy problem with a neural implicit representation that reconstructs the left and right ventricles, including the endocardial and epicardial boundaries of the LV myocardium. The study is closely related to the problem considered here because both approaches seek continuous cardiac shape from incomplete observations. Brás et al. use image data directly, whereas this thesis isolates the geometric problem by using segmented contour points.

Muffoletto et al. (2026) also reconstruct three-dimensional cardiac shape from sparse segmentations with a neural implicit representation. Their method uses a learned biventricular shape prior and includes both SAX and long-axis views. It is therefore broader than the LV-focused SDF reconstruction considered here. The study does not report a public dataset or code release, which limits its direct reproducibility and comparison with this work.

Two techniques are critical for applying INRs to thin cardiac structures where high-frequency geometric detail must be preserved. Fourier features reduce the low-frequency bias of standard multilayer perceptrons, and Eikonal regularisation encourages the learned field to behave like a signed distance function (Gropp et al., 2020; Tancik et al., 2020). These ideas support smooth interpolation without abandoning local surface detail.

Recent cardiac INRs also model motion across the cardiac and respiratory cycles (Kunz et al., 2024; Quillien et al., 2025; Tian et al., 2026; R. Wang et al., 2025). Their main goal is temporally coherent image or volume reconstruction. The present work has a narrower geometric goal: it conditions paired surface reconstruction on ED or ES and then measures the wall between those surfaces.

For encoding unordered input observations such as SAX contour points, the PointNet architecture (Qi et al., 2017) provides a permutation-invariant encoder via shared MLPs followed by symmetric max-pooling. This is a natural match for SAX contours because their point ordering should not change the reconstructed anatomy. A PointNet-style encoder can therefore reduce the observed contour set to a latent description used by an implicit decoder.

2.3.4. Wall-thickness measurement techniques

Myocardial wall thickness is a primary clinical biomarker for cardiac remodelling. Grossman et al. (1974) demonstrated that wall thickness is a principal determinant of left-ventricular diastolic stiffness, with hypertrophied ventricles exhibiting markedly elevated stiffness proportional to end-diastolic wall thickness. More recently, Huelnhagen et al. (2017) showed at 7.0T MRI that wall thickness correlates with tissue transverse relaxation properties, confirming its value as a non-invasive surrogate for myocardial tissue characterisation. The AHA 17-segment model (Cerqueira et al., 2002) provides the standard framework for reporting regional thickness values.

The accuracy of wall-thickness measurement depends critically on the quality of the underlying surface reconstruction, as any smoothing or geometric artefact propagates directly into the thickness estimate. The literature describes five families of measurement methods, each with distinct trade-offs.

The simplest approach uses nearest-neighbour distance: for each endocardial vertex, a KD-tree identifies the closest epicardial point, yielding a measurement in $O(N \log M)$ time. Its runtime depends on mesh size and the implementation. This method can provide a lower bound on the true transmural thickness because it ignores the anatomical direction of measurement — the nearest point may lie tangentially rather than across the wall.

Ray-based methods address this limitation by casting a ray from each endocardial vertex along its surface normal until it intersects the epicardium. The intersection is computed via the Möller–Trumbore algorithm, accelerated with a Bounding Volume Hierarchy (BVH) to $O(\log F)$ per query. This approach measures transmurally but is sensitive to normal noise: a slightly rotated normal can produce a spurious long or short ray. The Shape Diameter Function variant improves robustness by casting a cone of K rays (typically $K{=}7$) at a half-angle of $30°$ around the normal and reporting the median intersection distance, effectively filtering outlier hits caused by local mesh irregularities.

Volumetric methods operate on voxelised representations. The Euclidean Distance Transform (EDT) computes, for each myocardial voxel, the distance to the nearest boundary, and the sum of the endocardial and epicardial distance fields approximates local thickness ($t = D_{\text{endo}} + D_{\text{epi}}$). This estimate is exact for planar parallel boundaries but can be biased in curved regions because the nearest-boundary directions do not necessarily follow a common transmural path. The method is computationally efficient ($O(V)$ via the Saito–Toriwaki algorithm) but limited by voxel resolution.

Partial differential equation (PDE)-based methods are widely used for transmural thickness estimation. Jones et al. (2000) solved Laplace's equation between the two tissue boundaries and used the resulting non-intersecting field lines to establish corresponding paths. Thickness is obtained from the path length between the boundaries along these streamlines. Yezzi and Prince (2003) formulated an

Eulerian PDE method that computes distances propagated from both boundaries along the Laplacian field. Their sum gives the local tissue thickness.

Finally, mesh-based regularised correspondence methods combine an initial geometric estimate (normal projection or nearest neighbour) with iterative Laplacian smoothing on the mesh connectivity graph. At each iteration, each vertex's thickness is updated as a weighted average of its neighbours, anchored to the initial estimate. This produces spatially smooth thickness maps suitable for bullseye visualisation, at the cost of potentially attenuating real local thickness variation if the smoothing parameter is set too aggressively.

2.3.5. Summary and research gap

The reviewed literature shows that each component of the LV analysis pipeline is well-studied in isolation: deep learning gives accurate slice-wise segmentation, SSMs encode population-level shape priors, implicit representations provide continuous surface modelling, and multiple wall-thickness algorithms exist. To the best of our knowledge, however, limited work integrates these components into a single pipeline for the specific setting addressed in this thesis. The combination that appears to be missing is a method that encodes sparse SAX contour points, decodes paired endocardial and epicardial surfaces as implicit signed distance fields, conditions the reconstruction on cardiac phase, constrains the wall offset to stay positive, and trains jointly on real clinical data and SSM-derived synthetic samples. This is useful because the wall can then be measured everywhere on a continuous surface instead of only on the acquired slices. This gap motivates the approach described in the remaining chapters.

CHAPTER 3

Methodology

This chapter describes the methodology used to reconstruct 3D LV geometry from sparse SAX contours and to measure myocardial wall thickness on the reconstructed model. The proposed pipeline combines synthetic and real cardiac MRI-derived data, a phase-conditioned INR, and a systematic comparison of local wall-thickness algorithms.

3.1. Overview

The method is determined by three constraints. First, SAX contours provide detailed boundaries only on a small number of image planes. Second, clinical datasets do not provide independent 3D surfaces for direct supervision. Third, the endocardium and epicardium must remain ordered so that they define a valid myocardial wall. The pipeline addresses these constraints with a shared contour representation, synthetic pre-training followed by clinical fine-tuning, and a coupled implicit decoder with positive wall separation.

Section 3.2 describes how synthetic and clinical cases are placed in the common representation. Section 3.3 presents the contour encoder and implicit decoder, and Section 3.4 gives the optimisation and surface-extraction procedure. Finally, Section 3.5 defines the local estimators and AHA-17 aggregation applied to the reconstructed surfaces. The method evaluates geometric completion and computational measurement consistency; it does not establish independent anatomical truth.

Figure 3.1 summarises the complete workflow. Panels (a) and (b) use ACDC data, while panels (c)–(e) use examples from the UK Digital Heart Project LV SSM pipeline.

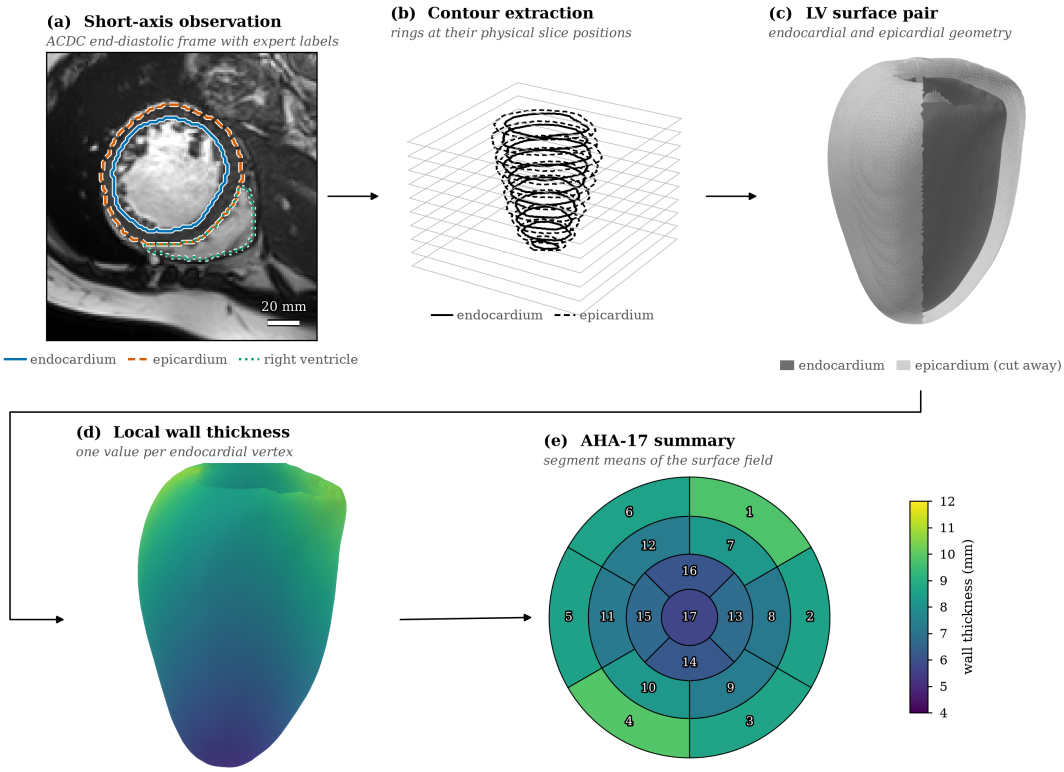

Figure 3.1. Preview of the end-to-end methodology from sparse SAX observations to reconstructed LV geometry and wall-thickness outputs. Panels (d) and (e) share one colour scale. Panels (c)–(e) illustrate the geometry and measurement stages using the SSM mean shape with a synthetic epicardial offset, so the thickness field shown there is illustrative rather than a reported result.

3.2. Data Preparation

A fundamental challenge in training the model is the absence of a paired dataset that provides both sparse SAX contour observations *and* independently acquired three-dimensional LV meshes. Clinical cardiac MRI datasets supply segmentation masks and image volumes, but they do not include the watertight endocardial and epicardial surfaces that an SDF decoder requires as supervision. This absence of independent three-dimensional targets is the starting point for the entire data-preparation strategy.

To address this issue, a two-stage approach was used. In the first stage, 1300 synthetic ED meshes were generated from the UK Digital Heart Project SSM of the LV, providing anatomically plausible three-dimensional geometry with controlled shape variation. The model was pre-trained on this synthetic corpus, which established a geometric prior for LV shape variability. In the second stage, the training corpus was extended with meshes reconstructed directly from real clinical segmentations: ED and ES frames from the ACDC and M&Ms-2 datasets. These meshes are segmentation-derived surrogate targets, rather than independent anatomical ground truth. This fine-tuning stage introduced patient-specific anatomy, acquisition variability, and phase diversity that the SSM alone cannot provide.

All three data streams were converted into the same cache schema. The shared format allows the pre-training and fine-tuning stages to use the same training loop and loss function without any dataset-specific logic.

3.2.1. Absence of independent 3D targets and motivation for SSM

The limitation outlined above can be stated more precisely as a supervision-gap problem: the model requires paired samples of sparse SAX contours and reliable three-dimensional LV meshes, while public cine MRI benchmarks provide mainly slice-wise labels. In the ACDC (Bernard et al., 2018) and M&Ms-2 datasets, the available annotations are segmentation masks rather than independently acquired watertight endocardial and epicardial surfaces suitable for direct SDF supervision.

Two complementary strategies were therefore adopted to create the missing supervision. First, an SSM was used to generate a large number of synthetic ED meshes that are anatomically valid by construction. This supplied geometric supervision for the pre-training stage at the cost of not capturing real segmentation artefacts or scanner-specific anatomy. Second, real ED and ES segmentation masks were converted into watertight three-dimensional meshes through a controlled preprocessing pipeline. This provided the patient-specific geometry needed for fine-tuning, at the cost of requiring careful mesh reconstruction from inherently discrete voxel data.

The SSM pre-training approach is motivated by the observation that learning the geometry of the LV, rather than fitting individual segmentations, requires exposure to the full range of normal and pathological LV shapes. The SSM encodes this range compactly as a mean shape and a set of principal modes of variation, and sampling from its latent space produces diverse but anatomically constrained meshes. The real-data fine-tuning stage then ensures that the model generalises to the segmentation quality and acquisition conditions present in the evaluation datasets.

3.2.2. Statistical Shape Model of the left ventricle

An SSM represents the variability of an anatomical structure across a population by applying principal component analysis (PCA) to a set of aligned training meshes (Heimann & Meinzer, 2009). Each mesh is expressed as a vector of vertex coordinates; the mean of all training shapes forms the mean shape $\bar{X}$, and the eigenvectors of the covariance matrix of the centred shapes form the principal modes. Any shape in the population can be approximated as a linear combination of these modes, and new shapes that follow the observed statistics can be generated by sampling the latent coefficient vector.

This thesis uses the UK Digital Heart Project LV SSM reported by Bai et al. (2015), built from more than 1,000 high-resolution cardiac MR images of healthy subjects. The model provides a mean endocardial surface of 22 043 vertices together with 100 principal modes of variation stored as a $66\,129 \times 100$ matrix of mode vectors and a corresponding vector of 100 eigenvalues λ_i. The standard deviations $\sigma_i = \sqrt{\lambda_i}$ quantify the magnitude of each mode. A new endocardial mesh is generated by

$$X(b) = \bar{X} + \Phi b, \tag{3.1}$$

where $\bar{X}$ is the mean shape flattened to a vector, Φ is the matrix of mode vectors, and $b = [b_1, \ldots, b_{100}]^\top$ is the sampled coefficient vector.

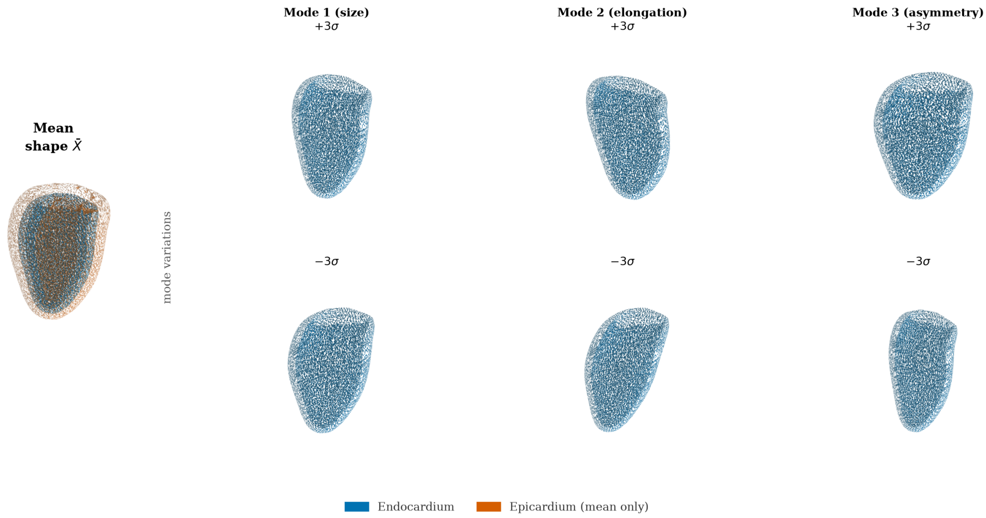

Figure 3.2. UK Digital Heart Project LV SSM mean shape and first three PCA modes at $\pm 3\sigma$.

The first few modes capture the largest sources of shape variation in the population; mode 1 primarily controls overall LV size, mode 2 encodes elongation along the long axis, and higher modes represent progressively more localised deformations, as shown in Figure 3.2.

3.2.3. Synthetic end-diastolic dataset (SSM pre-training)

The synthetic ED dataset was built by sampling 1300 meshes from the SSM and converting each mesh into a training-compatible sample. The generation pipeline proceeds in five steps, illustrated in Figure 3.3.

Step 1: Latent coefficient sampling. Each candidate coefficient vector b was drawn by sampling each component as $b_i \sim \mathcal{N}(0, \sigma_i^2)$ and then applying two acceptance criteria:

$$\left|\frac{b_i}{\sigma_i}\right| \leq 3, \qquad \sum_i \left(\frac{b_i}{\sigma_i}\right)^2 \leq \chi^2_{0.99,d}, \tag{3.2}$$

where σ_i is the standard deviation of mode i, d is the number of active modes, and $\chi^2_{0.99,d}$ is the 99th percentile of the chi-squared distribution with d degrees of freedom. The per-mode clipping prevents individual modes from dominating the shape, and the Mahalanobis bound prevents the combined deformation from falling outside the plausible population distribution. Accepted samples are scattered within a bounded ellipsoid in latent space, as shown in panel (c) of Figure 3.3.

Step 2: Mesh generation. Each accepted coefficient vector b is substituted into Equation (3.1) to produce a deformed endocardial vertex array, which is then triangulated using the face connectivity of the mean-shape mesh. The resulting triangular surface is watertight and oriented by construction.

Step 3: Quality filtering. Before a generated mesh is accepted into the training set, it is evaluated against four proxy metrics: enclosed volume (mL), surface area (cm^2), sphericity index, and plausibility of the apical and basal normal directions. Samples that fall outside calibrated bounds for any of these metrics are discarded. Periodic exact audits using VTK mass-properties computation were also performed every 40 samples to verify the proxy metric calibration.

Step 4: Synthetic epicardium. The SSM provides only the endocardial surface. The epicardial surface was synthesised by offsetting each endocardial vertex p_i along its outward normal n_i by a spatially varying distance d_i,

$$q_i = p_i + d_i n_i, \tag{3.3}$$

where d_i was set to approximately 10 mm near the base and 5 mm near the apex, with additive Gaussian noise of standard deviation 0.5 mm to avoid perfectly uniform geometry. This synthetic epicardium is a geometric pre-training prior only. It is not intended to reproduce clinical wall thickness, and it is never used as a reference for any wall-thickness result. Its purpose is to give the decoder a consistent outer surface during pre-training, so that the model learns the general shape relation between an inner and an outer LV surface before it sees real data.

Step 5: SAX contour extraction and volumetric supervision points. Each pair of endocardial and epicardial surfaces was sliced at 10 evenly spaced axial levels to produce sparse SAX contour rings, matching the acquisition pattern of real cine MRI. For each case, 2048 query points were sampled near the surfaces and in the surrounding volume, and signed-distance targets or inside/outside labels were computed. These query points form the dense volumetric supervision used by the SDF loss terms during training.

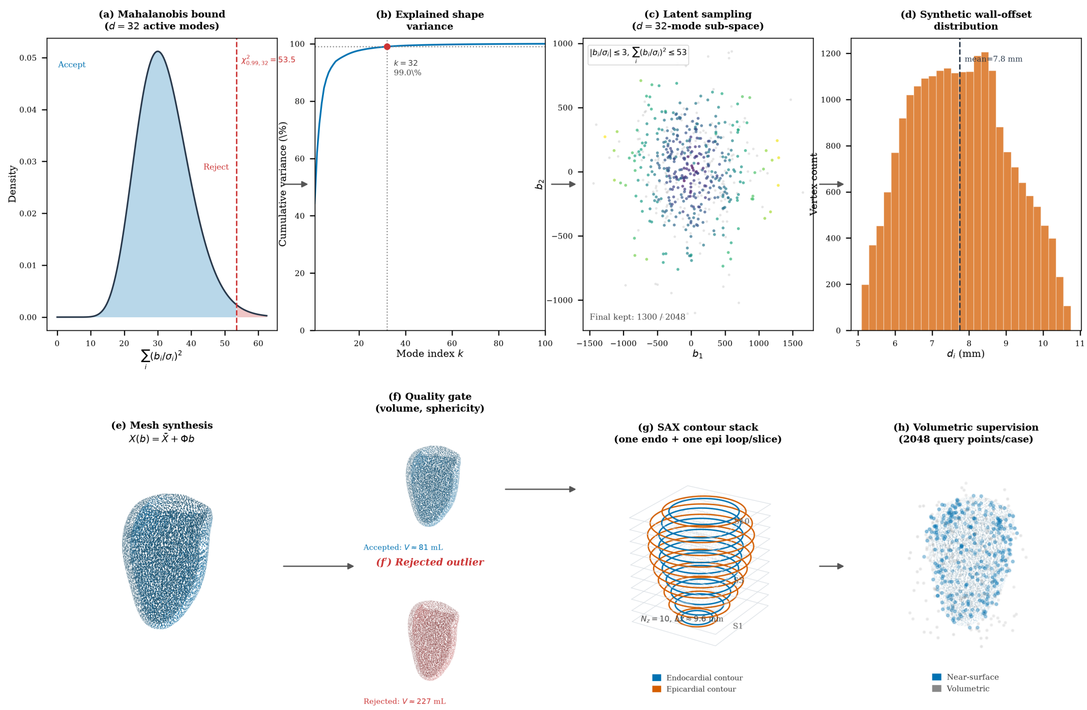

Figure 3.3. Synthetic ED dataset generation from SSM sampling to geometric supervision. Top row: statistical constraints; bottom row: geometric construction and supervision targets.

Figure 3.3 summarises this process in two rows. The top row provides the statistical view of sampling: panel (a) shows the Mahalanobis-statistic density and the $\chi^2_{0.99,32}$ acceptance threshold, panel (b) shows that 32 active modes capture 99% cumulative shape variance, panel (c) shows accepted latent samples in the first two dimensions, and panel (d) shows the synthetic epicardial-offset distribution from

Equation (3.3). The bottom row provides the geometric view: panel (e) shows mesh synthesis via Equation (3.1), panel (f) contrasts an accepted shape with a rejected outlier, panel (g) shows the 10 SAX contour levels, and panel (h) shows the 2048 near-surface and volumetric query points used for decoder supervision.

These fixed SSM-sampling and supervision settings were kept constant across all synthetic ED cases to ensure reproducible pre-training before fine-tuning on real ED and ES data.

3.2.4. Real ED and ES labels: from SAX segmentation to 3D supervision

SSM-derived meshes alone cannot capture the full diversity of clinical acquisition conditions: scanner differences, segmentation quality, image resolution, and the distinct geometry of the end-systolic phase. A second data stream was therefore created by reconstructing watertight three-dimensional LV meshes directly from the segmentation masks provided in the ACDC (Bernard et al., 2018) and M&Ms-2 datasets, taking both the ED and ES frames so that the model is exposed to both cardiac phases.

These real ED and ES caches are used in the second optimisation stage, after the initial SSM-based training. The reason for this fine-tuning stage is that the synthetic corpus teaches a strong global LV-shape prior, but it does not fully reproduce clinical variability in contour quality, inter-dataset label style, and phase-specific contraction patterns. Fine-tuning on real ED and ES therefore reduces domain shift between synthetic supervision and the target clinical setting while preserving the geometric regularity learned from the SSM.

Operationally, the fine-tuning stage keeps the same model architecture and the same multi-term loss from Section 3.4, but replaces the synthetic-only stream with real-derived cache samples from ED and ES frames. Because the real caches follow the same schema (contours, tissue labels, phase label, mesh supervision, and query points), the training loop can continue from the pre-trained checkpoint without changing model interfaces. The phase label then provides the conditioning signal that allows one shared model to adapt from diastolic to systolic geometry during continued training.

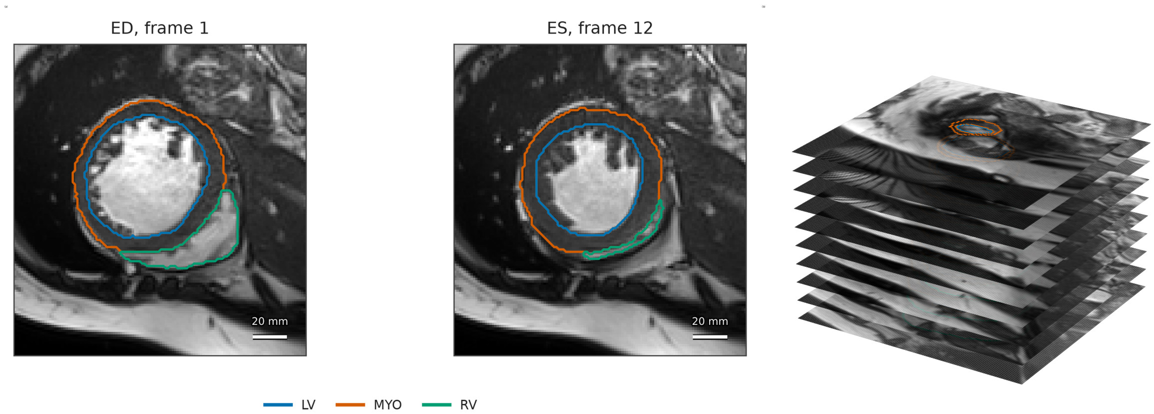

Figure 3.4. Clinical-input view used in the real-data pipeline: (a) representative ACDC ED/ES SAX slices with labels and (b) the corresponding sparse ED SAX stack in physical three-dimensional spacing.

Before describing how these meshes are built, it is useful to look at the raw clinical input, because it makes clear why a reconstruction step is needed at all. Figure 3.4 shows representative SAX slices from the ACDC dataset (Bernard et al., 2018), with the LV cavity, myocardium, and right-ventricular contours overlaid. Each frame is only a two-dimensional cross-section of the heart, and the segmentation is defined one slice at a time. Panel (b) of Figure 3.4 stacks the segmented ED slices at their physical through-plane positions and exposes the central difficulty of the problem: a clinical acquisition provides only a small number of parallel SAX slices separated by several millimetres, so the observed geometry is a set of sparse contour rings rather than a continuous surface.

The reconstruction pipeline closes this gap by converting each sparse label stack into a derived, watertight three-dimensional mesh. These meshes are *segmentation-derived surrogate targets*, not manually annotated or independently acquired 3D reference geometry: their geometry is determined by the observed labels and by the explicitly stated resampling, smoothing, and capping operations. This distinction is important when interpreting the later reconstruction-consistency results. An earlier ES pipeline attempted to fit the SSM to systolic contours; this was abandoned because forcing a diastolic statistical prior onto systolic geometry produced anatomically unrealistic surfaces with collapsed walls. The direct reconstruction approach instead retains the patient-specific geometry present in the ED or ES labels. The two parallel endocardial and epicardial paths are shown in Figure 3.5, and the four stages of the conversion are described below. Table A.2 of Appendix A lists the individual operations in order.

From label volume to clean binary masks. Each segmentation volume is read from NIfTI format and reoriented to the RAS+ convention using its affine, which places the voxel axes in a consistent patient frame while preserving the physical spacing. This step does not define the cardiac long axis. The acquired SAX stack supplies the through-plane direction, and base and apex are identified later from the cross-sectional areas at the two ends of the stack. The ACDC and M&Ms-2 datasets use compatible but not identical label encodings for the LV cavity, the myocardium, and the right ventricle, so each source is first mapped to a common convention. Two binary masks are then formed. The endocardial mask is the LV cavity label, and the epicardial mask is the union of the cavity and the myocardium:

$$M_{\text{endo}} = \mathbb{1}[L = L_{\text{LV}}], \qquad M_{\text{epi}} = \mathbb{1}[L \in \{L_{\text{LV}}, L_{\text{MYO}}\}], \tag{3.4}$$

where L is the label image, L_{LV} is the LV cavity label, and L_{MYO} is the myocardium label. The volume is then resampled to 1.5 mm isotropic spacing with nearest-neighbour interpolation, which avoids blending one label into another. Within each slice, the largest connected component of each label is kept and any holes are filled, and the largest three-dimensional component is kept afterwards. This removes the small disconnected fragments that often appear near the first and last slices of the stack. The endocardial mask is finally merged into the epicardial one so that $M_{\text{endo}} \subseteq M_{\text{epi}}$ holds everywhere.

From masks to a smooth closed surface. Applying marching cubes directly to voxel labels produces a visible staircase, so the binary mask is first smoothed with a Gaussian filter at a scale of 1.5 mm and the surface is taken at its 0.5 isosurface (Lorensen & Cline, 1987). Panels (d) and (d′) of Figure 3.5 show the unsmoothed voxel surface for the endocardial and epicardial masks, which makes this artefact explicit. Taubin λ/μ smoothing (Taubin, 1995) (25 iterations, $\lambda = 0.53$, $\mu = -0.55$) then removes most of the remaining staircase while approximately preserving the enclosed volume. Because the labels stop at the valve plane, marching cubes closes the top of the ventricle with a rounded dome that does not exist in the anatomy. The upper 6% of the apex-to-base extent is therefore cut away, the opening is closed with a flat cap perpendicular to the long axis, and three iterations of cotangent-Laplacian fairing smooth the join. Quadric-error decimation (Garland & Heckbert, 1997) finally reduces every mesh to at most 4000 vertices, so that all cases reach the cache at a comparable size.

Plausibility checks. Marching cubes reproduces whatever surface the labels imply, even when that surface is not a plausible ventricle. A label stack can be incomplete, noisy, or come from a long-axis rather than a short-axis acquisition, and the resulting mesh is then valid geometry but poor anatomy. Seven checks are therefore applied before a mesh enters the cache, and a case that fails any of them is discarded, because training on implausible geometry would weaken the shape prior the model is meant to learn.

Three of the checks ask whether the stack really covers the ventricle. The cross-sectional area of the endocardium must decrease from the mid-cavity towards the apex, because the ventricle narrows in that direction; when the apical end is the wider one, base and apex have most likely been swapped, or the stack stops before the true apex. The most basal slice must have an area consistent with the valve-plane region, since a very small basal contour means the acquisition ended too early and the basal cap would then close the mesh at an arbitrary height. The ratio between the apex-to-base length and the largest short-axis diameter must stay inside a fixed range, which removes globally distorted shapes.

The remaining four checks ask whether the segmentation is internally consistent. The slice centroids must not jump sideways from one slice to the next, because that pattern points to motion or an orientation error and produces twisted surfaces. Within each slice, the endocardial and epicardial rings must stay smooth and roughly convex, since jagged or self-intersecting rings usually come from local segmentation failures that smoothing cannot repair. At every level the endocardium must lie inside the epicardium with a positive gap between them, because the opposite ordering is an impossible target for a decoder that models the wall as an outward offset. The average gap between the two rings must also stay within plausible limits, which removes cases where labels have merged or leaked into neighbouring tissue.

Contour and query-point extraction. The encoder receives contours extracted directly from the acquired clinical slices, not contours re-sliced from the derived mesh. For each available SAX slice, the largest endocardial and epicardial label contours are retained and resampled to 60 evenly spaced points. Their physical slice positions are kept, so the encoder receives the original sparse observation in its measured through-plane arrangement. This preserves the acquisition geometry and prevents the derived mesh from leaking into the encoder input.

The derived endocardial and epicardial meshes are centred and normalised with the same translation and scale before they are used as three-dimensional supervision. For each mesh pair, 2048 query points are drawn near the two surfaces and across a padded bounding box. Near-surface points teach the decoder where the surfaces lie; the volume points teach whether a location is inside or outside each surface. Every query point receives occupancy labels for both surfaces. The contours are therefore the sparse input from the clinical slices, whereas the query points and mesh samples are training-only targets derived from the mesh.

Figure 3.5 provides a visual summary of the conversion from clinical labels to mesh-ready supervision.

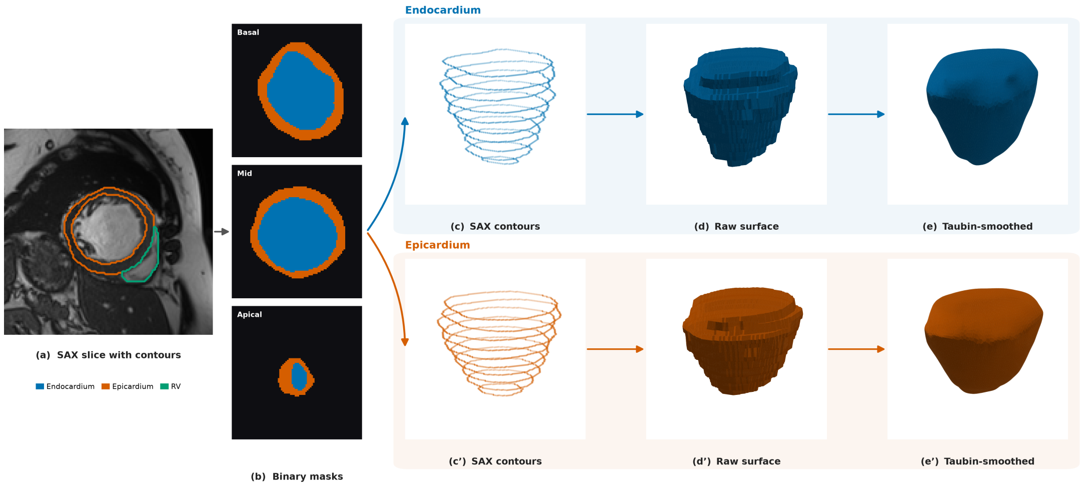

Figure 3.5. Real ED clinical-label-to-mesh pipeline (ACDC): from SAX labels and binary masks to endocardial and epicardial surface branches used for derived supervision.

Panel (a) shows the acquired SAX image with LV cavity, myocardium, and RV labels, and panel (b) shows the corresponding binary endocardial and epicardial masks after label harmonisation. The pipeline then branches into separate endocardial and epicardial paths: panels (c)–(e) and panels (c′)–(e′), respectively.

In both branches, the voxel surface view makes discretisation artefacts explicit, and Taubin smoothing illustrates their reduction before later post-processing. The figure is intentionally a process overview; the full sequence also includes basal capping, cap fairing, decimation, and the plausibility checks. Contours for the encoder remain those extracted from the original clinical slices.

3.2.5. Shared cache representation and contour augmentation

All cases are stored in a common cache representation so that the model receives the same interface regardless of whether a sample originates from the SSM or from a clinical segmentation.

A cache item combines the sparse contour observation with mesh-derived supervision and the metadata needed to keep both representations in a shared coordinate frame. The ED and ES caches contain 10 SAX levels, 60 points per available contour ring, and 2048 query points per case. This design deliberately separates what the model observes from what it is allowed to learn from: the encoder receives only contour coordinates, tissue identity, and cardiac phase, whereas the decoder is supervised by surface and volumetric samples derived from the corresponding mesh. Thus, at inference time, the same model can reconstruct geometry from sparse contour observations alone.

Table 3.1. Fields of one shared cache item and their role in training.

Field	Origin	Role
Contour points	Synthetic mesh slicing or clinical slice labels	Sparse SAX observation for the encoder and contour-anchor loss
Tissue label	Contour extraction	Distinguishes endocardial from epicardial points during tissue-specific pooling
Phase label	Dataset metadata	Conditions the encoder on ED or ES geometry
Surface samples	Prepared endocardial and epicardial meshes	Surface-fitting and normal-alignment supervision
Query points and targets	Near-surface and volumetric mesh sampling	Dense SDF, occupancy, sign, and exterior supervision
Mesh metadata	Preprocessing pipeline	Coordinates, scale, faces, and orientation for evaluation and visualisation

Table 3.1 makes this separation explicit. The first three fields describe the clinical-style observation supplied to the encoder; the remaining fields are available only during training, evaluation, or visualisation. Keeping these roles distinct prevents mesh-derived information from leaking into the encoder input while allowing dense supervision of the implicit field.

This common contract enables two-stage optimisation without dataset-specific changes to the architecture or loss. The model is first trained on synthetic ED samples, which establish a broad geometric prior, and is then fine-tuned on real ED and ES samples, which introduce patient-specific anatomy and phase-dependent shape variation. The loader changes only the source stream between stages; the fields consumed by the training loop remain unchanged.

Contour augmentation is applied to both synthetic ED and real ED/ES training observations, after the train/validation/test split. The target mesh, surface samples, query points, occupancy labels, tissue labels, and phase label remain fixed. This is therefore an observation-robustness intervention rather than the creation of a geometrically transformed anatomical target. Such label-preserving perturbations are commonly used to improve robustness when labelled data are limited (Chlap et al., 2021; Shorten & Khoshgoftaar, 2019).

For each augmented sample, the loader draws small independent in-plane translations per slice, point-wise in-plane jitter, global scale jitter, and contour-point dropout. Slice dropout and long-axis rotation are applied probabilistically while retaining at least three slice levels. The endocardial and epicardial contours within a slice receive the same translation, preserving their local relationship. As illustrated in Figure 3.6, these perturbations ask the encoder to recover one fixed mesh-derived target from plausible variations of its sparse SAX observation.

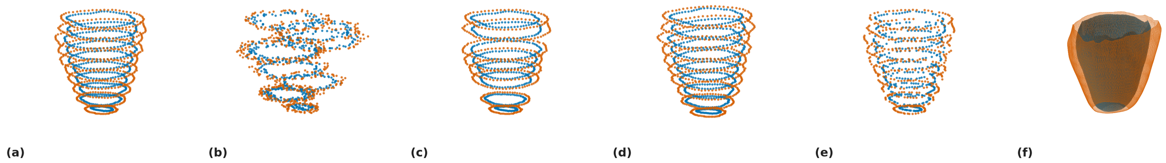

Figure 3.6. Examples of contour-input augmentation. Endocardial and epicardial points are shown in blue and orange, respectively.

In Figure 3.6, panel (a) shows the unaltered contour stack, while panels (b), (d), and (e) show slice translation, slice rotation, and contour-point dropout, respectively. Panel (f) shows the corresponding 3D mesh label. Thus, the sparse SAX observation changes across the augmented examples, whereas the derived mesh and all decoder-supervision targets remain fixed; the figure illustrates robustness of the encoder input rather than transformation of the anatomical ground-truth geometry.

3.2.6. Dataset composition and splits

This subsection reports how many samples each data stream finally contributes and how the real streams are divided for training and evaluation. The real data come from two public collections: the Automated Cardiac Diagnosis Challenge (ACDC) and the second Multi-Centre, Multi-Vendor and Multi-Disease Cardiac Segmentation challenge (M&Ms-2). Together they provide 460 patients (100 from ACDC and 360 from M&Ms-2). For each patient we reconstruct one ED mesh and one ES mesh, giving two real streams. Both collections are openly available for research at no cost; each requires a free registration and the acceptance of a data-use agreement. The original M&Ms challenge paper (Campello et al., 2021), discussed in Chapter 2, describes a separate 375-study dataset and is not used as a citation for the M&Ms-2 release. The synthetic stream is generated from the UK Digital Heart Project SSM: of its 100 modes, the first 32 already capture 99% of the shape variance, so we sample within that sub-space. Drawing shape coefficients under the Mahalanobis bound of Equation (3.2) and rejecting implausible shapes yields 1300 accepted meshes out of 2048 candidates (63%). The shape model is likewise open and free: it is distributed as a public repository by the UK Digital Heart Project,[1] needs no access application, and the authors ask only that the atlas papers be cited. None of the three sources used in this work therefore required a licence fee.

Not every real patient produces a usable mesh. A segmentation may be a long-axis view, cover too few slices, or fail an anatomical plausibility check, such as an endocardium that is not fully enclosed by the epicardium. After these quality gates, 444 of the 460 patients yield a valid ED mesh and 423 yield a valid ES mesh. The surviving meshes are then divided for training and evaluation. To prevent information from leaking between the partitions, the real streams are split by patient rather than by mesh, so that all meshes from a given patient stay within a single partition. Both streams use the same 70 %/15 %/15 %

[1] https://github.com/UK-Digital-Heart-Project/Statistical-Shape-Model

allocation for training, validation, and test, rounded to whole patients, and the synthetic ED meshes are used only to enrich the pre-training stage and are never placed in the validation or test sets.

Table 3.2. Final sample counts and partitioning of the three data streams. Real streams report patients that pass the quality gates; the synthetic stream reports meshes accepted out of the 2048 sampled shape vectors. The reported clinical evaluation contains no synthetic cases.

Stream	Source	Phase	Count
Real data	ACDC and M&Ms-2	ED	444 patients
Real data	ACDC and M&Ms-2	ES	423 patients
Synthetic data	UK Digital Heart SSM	ED	1300 meshes

Table 3.2 records the two real source collections and the synthetic pre-training stream. The role of each stream in the two-stage training procedure is summarised in Appendix A.

3.3. Model Architecture

The model is built from two complementary parts. The encoder reads the sparse contour points and summarises the case into a compact latent representation, while a local contour-volume branch keeps the surrounding spatial structure in a coarse 3D feature field. The decoder then combines the global latent code, the local feature at each query point, and the point coordinates to predict the LV endocardial signed-distance field. Because the decoder can be evaluated anywhere in space, the reconstructed surface is not tied to a fixed grid or a fixed number of vertices, and a mesh is only generated once the field has been sampled densely. The output is then a continuous signed-distance representation of the chamber and its surrounding wall.

This design matches the structure of the problem. The contour input is a small, unordered set of points, so the encoder uses a point-cloud formulation that is invariant to the order of the points. The output is a smooth, continuous surface, so the decoder is an implicit function that can be evaluated at any resolution. The epicardial field is defined through a positive offset from the endocardial field, preventing collapse of the two predicted fields. Anatomical wall thickness is subsequently measured between the extracted surfaces.

Figure 3.7 links the global shape code, local feature volume, and coupled decoder. The offset shown in the figure is a separation between implicit fields, not a measured transmural distance. The encoder is formalised in Section 3.3.1 and the decoder in Section 3.3.2.

3.3.1. Contour encoder

The encoder receives, for each contour point, its 3D coordinates together with a label indicating whether the point is endocardial or epicardial and a label indicating the cardiac phase. A small shared network processes every point independently and turns it into a feature vector. The features of the endocardial points and the features of the epicardial points are then combined separately by taking, for each feature, its maximum over the points. This makes the latent code invariant to the order of the contour points, which is important because contour points have no natural ordering across patients or slices. The two resulting descriptors are joined and projected to a single 256-dimensional latent vector z. At the same time,

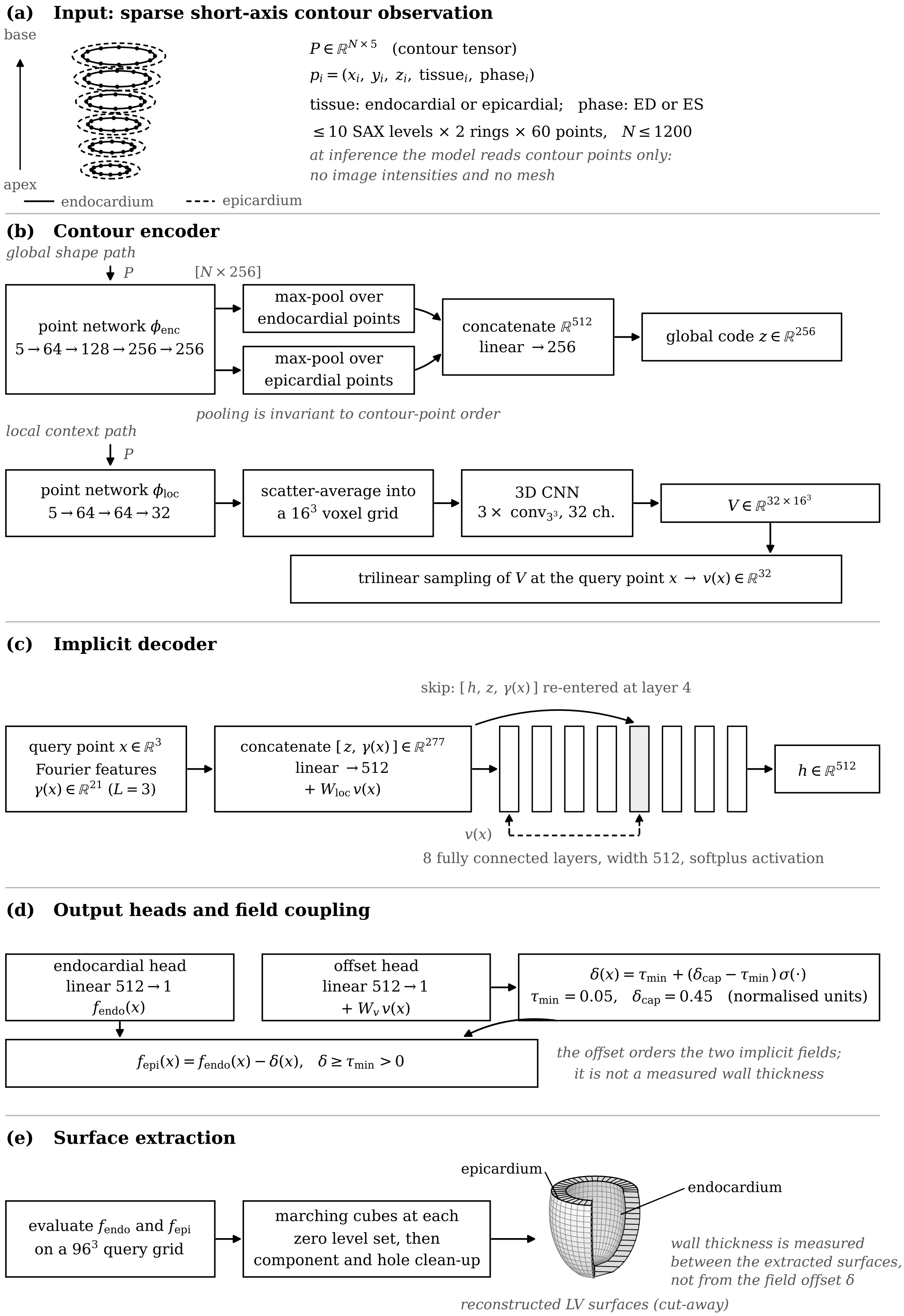

Figure 3.7. Phase-conditioned model architecture. Tissue-specific global and local contour features condition an implicit decoder, which predicts the endocardial field and a positive wall offset before mesh extraction.

the same point features are scattered into a local 3D feature volume that keeps spatial neighbourhood information around the LV wall.

More formally, let $P = \{p_i\}_{i=1}^{N}$ be the contour points of one case, where each p_i carries coordinates, a tissue label, and a phase label. The shared point network computes

$$h_i = \phi_{\text{enc}}(p_i), \tag{3.5}$$

where ϕ_{enc} is the shared network. The endocardial and epicardial descriptors are then obtained by max-pooling within each tissue,

$$z_{\text{endo}} = \max_{i \in \mathcal{E}} h_i, \qquad z_{\text{epi}} = \max_{i \in \mathcal{P}} h_i, \tag{3.6}$$

where $\mathcal{E}$ and $\mathcal{P}$ are the endocardial and epicardial point sets. The global latent code is then

$$z = W_z[z_{\text{endo}}, z_{\text{epi}}] + b_z, \tag{3.7}$$

while the same scattered features are aggregated into a local 3D volume V and sampled at each query point by trilinear interpolation,

$$v(x) = \text{Interp}(V, x). \tag{3.8}$$

This local field adds spatial detail to the decoder while the global latent vector keeps the overall chamber shape compact and stable.

Separate pooling preserves the distinct roles of the cavity and outer wall, while the local volume supplies spatial context near each query point.

3.3.2. Implicit decoder and positive-wall coupling

Before a query point is given to the decoder, its coordinates are expanded with a Fourier positional encoding. This simply adds a few sine and cosine copies of the coordinates at different frequencies, which helps the network represent fine geometric detail instead of only smooth, blurry shapes (Tancik et al., 2020). Three frequency bands are used. For a query point $x \in \mathbb{R}^3$ the encoding is

$$\gamma(x) = [x, \sin(2^0\pi x), \cos(2^0\pi x), \ldots, \sin(2^{L-1}\pi x), \cos(2^{L-1}\pi x)], \tag{3.9}$$

with $L = 3$. The decoder itself is an eight-layer fully connected network of width 512 with a skip connection at the middle layer. It uses a smooth activation (softplus) so that the reconstructed surface is smooth and free of the sharp creases that a ReLU activation would introduce.

The decoder produces two outputs. The first is the signed distance to the endocardium, f_{endo}. The second is a positive decoder wall-offset field δ, which controls the separation of the predicted epicardial and endocardial fields at each query point. It is a learned difference between field outputs, not a Euclidean transmural distance measured between the extracted surfaces. The decoder also receives the local contour feature $v(x)$ from the spatial volume, which is injected into the model so that it can use local wall context when predicting the offset. The offset is squeezed into a fixed positive range by a sigmoid,

$$\delta(x, z, v) = \tau_{\text{min}} + (\delta_{\text{cap}} - \tau_{\text{min}})\sigma\left(g_\delta(x, z, v)\right), \tag{3.10}$$

where g_δ is the raw decoder output, σ is the sigmoid, and τ_{min} and δ_{cap} are the lower and upper bounds of the decoder wall-offset field. The epicardial field is then

$$f_{\text{epi}}(x, z, v) = f_{\text{endo}}(x, z, v) - \delta(x, z, v). \tag{3.11}$$

Because Equation (3.10) always keeps $\delta \geq \tau_{\text{min}} > 0$, the field offset cannot be zero or negative. This prevents a collapsed wall in the field, but it does not establish wall thickness, anatomical correctness, or the topology of the extracted meshes after discretisation. In the final configuration $\tau_{\text{min}} = 0.05$ and $\delta_{\text{cap}} = 0.45$ in normalised units, which corresponds to roughly 1.25–11.25 mm once mapped back to millimetres.

Table 3.3. Main architecture settings of the model.

Component	Configuration
Input features	3D coordinates, tissue label, and cardiac phase
Global encoder type	Shared network with endocardial and epicardial max-pooling
Local conditioning	Coarse 3D contour feature volume with trilinear interpolation
Latent dimension	256
Local volume size	16^3 with 32 channels
Positional encoding	Fourier features with $L = 3$ frequency bands
Decoder	8-layer network, width 512, skip connection at layer 4
Activation	Softplus (smooth)
Local injections	Local feature paths into the decoder trunk and offset head
Output heads	Endocardial signed distance and positive decoder wall-offset field
Offset range	$\tau_{\text{min}} = 0.05$, $\delta_{\text{cap}} = 0.45$ in normalised units
Inference grid	96^3 query grid for marching cubes

3.4. Training and Inference

This section explains how the model is trained from the shared cache and how a surface mesh, decoder offset map, and subsequent wall-thickness measurements are produced once training is finished. Training proceeds in two stages: the model is first pre-trained on the synthetic SSM cases and then fine-tuned on the real ED and ES cases. In both stages it must learn two things together: a valid distance field for the endocardium, and a sensible relationship between the endocardium and the epicardium.

3.4.1. Training data mix and augmentation

Training follows the two-stage strategy introduced in Section 3.2. In the first stage, the model is pre-trained only on the synthetic ED cases generated from the SSM. This large, controlled corpus teaches a strong prior for LV shape and tolerates aggressive augmentation. In the second stage, the pre-trained model is fine-tuned on the real cases reconstructed from ACDC and M&Ms-2, covering both the ED and ES phases. Fine-tuning starts from the pre-trained weights and continues with the same architecture and loss, so the model keeps the geometric regularity learned from the SSM while adapting to real segmentation quality and to the systolic phase. Throughout both stages the phase label is provided as input, so a single model handles ED and ES. Contour augmentation is used for both synthetic ED and real ED/ES observations; their surrogate targets and cached supervision are never changed.

A single phase-conditioned model is used instead of two separate models. The main reason is practical. When the experiments reported here were carried out, the public repository of the UK Digital Heart Project contained only the end-diastolic shape model, so a separate end-systolic model could not have benefited from the synthetic pre-training at all. An end-systolic shape model has since been added to that repository, but only after the training described in this chapter had been completed. Sharing one model lets the end-systolic cases reuse the shape prior learned from the synthetic end-diastolic corpus, and the phase label then tells the encoder which of the two configurations it is looking at. The cost of this choice is that the phase information is coarse. The model receives a binary label, not a continuous position in the cardiac cycle, so it can only represent two fixed states rather than the gradual change between them. Since the synthetic corpus is end-diastolic, the prior is also biased towards diastolic shape. Both effects make the end-systolic reconstruction the harder of the two cases, and this is visible in the results of Chapter 4.

The two stages are complementary. The synthetic ED cases give a large set of anatomically valid shapes but do not reproduce every artefact seen in real segmentations, which is why they are used for pre-training rather than as the final training data. Their role is to provide a geometric prior: they regularise the shape space, expose the model to a wide range of plausible LV geometries, and do so under controlled conditions. They do not provide realistic patient-specific wall thickness, because the epicardium in those samples is constructed by an offset rule. The real ED and ES cases then supply patient-specific anatomy from several datasets and let the model learn how the shape changes between the two phases. To avoid information leaking between the training, validation, and test sets, the real cases are split by patient before augmentation. Because augmentation perturbs only the observed contours, each real surrogate target remains tied to the anatomy of its own patient.

In practice, this means the encoder is trained to be robust to imperfect, noisy inputs, such as small shifts, jitter, missing slices, or dropped points, while the decoder is always supervised against a fixed, consistent 3D geometry.

3.4.2. Loss function

The training objective contains five complementary terms. Surface and dense SDF supervision fit the geometry, Eikonal regularisation preserves a valid distance field, off-surface repulsion suppresses spurious zero crossings, and direct wall-thickness supervision constrains the myocardial wall. The full objective is

$$\mathcal{L}_{\text{total}} = \lambda_{\text{surf}}\mathcal{L}_{\text{surf}} + \lambda_{\text{SDF}}\mathcal{L}_{\text{SDF}} + \lambda_{\text{eik}}\mathcal{L}_{\text{eik}} + \lambda_{\text{off}}\mathcal{L}_{\text{off}} + \lambda_{\text{WT}}\mathcal{L}_{\text{WT}}. \tag{3.12}$$

The surface term places the two zero-level sets on their target meshes. The SDF term provides dense volumetric supervision at cached query points. The Eikonal term encourages the field gradient to have unit magnitude (Gropp et al., 2020), while off-surface repulsion keeps free-space samples away from either zero-level set. Finally, the wall-thickness term directly supervises the predicted offset δ at endocardial surface points. The Eikonal term is computed in full precision to keep its gradient calculation numerically stable.

Table 3.4. Purpose of the five training-loss terms.

Term	Purpose	Effect on reconstruction
$\mathcal{L}_{\text{surf}}$	Fits zero-level sets to surface samples	Places endocardial and epicardial surfaces on the target meshes
$\mathcal{L}_{\text{SDF}}$	Fits cached query-point distances	Provides dense volumetric supervision
$\mathcal{L}_{\text{eik}}$	Encourages $\|\nabla f\| = 1$	Regularises the decoder as a signed-distance field
$\mathcal{L}_{\text{off}}$	Repels off-surface points from zero	Reduces spurious zero crossings away from the myocardium
$\mathcal{L}_{\text{WT}}$	Fits the predicted wall offset to wall targets	Constrains separation of the predicted fields

3.4.3. Optimisation and hardware

The retained training run uses AdamW with a learning rate of 2×10^{-5} and weight decay 5×10^{-4}. Its learning rate follows cosine annealing with warm restarts, with an initial restart period of 100 epochs. The synthetic pre-training stage lasted approximately 200 epochs; the recorded real-data fine-tuning history then continued for 776 epochs, from which the checkpoint at epoch 695 was selected by validation loss. Thus, a 400-epoch cap was not used for the final reported run. Fine-tuning resumes from the pre-trained weights rather than starting from scratch. Training uses mixed precision and gradient clipping at 1.0. Each of the two GPUs processes a local batch of 8 cases and gradients are accumulated for two steps, giving an effective batch size of 32. A light weight-clipping strategy was used instead of the more expensive spectral normalisation, since the decoder is evaluated at many points per step and GPU memory is the main constraint.

Table 3.5. Training configuration used for the model.

Setting	Value
Training strategy	SSM pre-training, then fine-tuning on real ED/ES
Training mode	Combined ED and ES with phase conditioning
Augmentation	Contour-input perturbations for synthetic ED and real ED/ES; targets unchanged
Optimiser	AdamW
Learning rate	2×10^{-5}
Scheduler	Cosine annealing with warm restarts ($T_0 = 100$ epochs)
Weight decay	5×10^{-4}
Training duration	Approximately 200 pre-training epochs, then 776 fine-tuning epochs (best checkpoint: 695)
Batch size	8 per GPU $\times$ 2 GPUs, accumulated for 2 steps (effective batch size 32)
Gradient clipping	1.0
Precision	Automatic mixed precision
Hardware	2 GPUs in parallel
Inference grid	96^3

3.4.4. Inference and decoder-offset extraction

At inference time, the model turns the input contours into the latent vector z and then evaluates the decoder on a regular 96^3 grid of points. Marching cubes reads the zero-level sets of f_{endo} and f_{epi} from this grid and produces the raw endocardial and epicardial meshes (Lorensen & Cline, 1987). For evaluation, degenerate faces are removed, the largest connected component is retained, holes are filled, and a mesh is remade from a cleaned occupancy mask when needed. This post-processing makes the evaluation geometry usable for surface and thickness measurements; it is not a guarantee provided by the neural architecture, and the reported metrics do not describe an entirely unmodified decoder output. A raw analytic offset is available at every endocardial vertex because the decoder predicts δ there. Converting this uncalibrated offset to millimetres gives

$$o_{\text{raw}}(x) = s\,\delta(x, z), \tag{3.13}$$

where s is the scale that maps normalised coordinates back to millimetres. This raw analytic offset is not measured after the fact by comparing two meshes, and it is not necessarily a Euclidean transmural separation. For visualisation, it is separately calibrated per phase by $o_{\text{cal}} = \alpha o_{\text{raw}}$, where α is the ratio between the mean segmentation-distance-transform reference and the mean raw analytic offset. This calibration fixes the mean magnitude by construction and is not an independent test of anatomical accuracy. Wall thickness is instead estimated on the reconstructed meshes by the methods in Section 3.5; the method judged most consistent in the comparison provides the final regional measurements.

Matched contour-lofting baseline. The reconstruction is also compared with a simple non-neural baseline built from exactly the same SAX contour rings. Each ring is represented by 60 points. The starting point of each successive ring is shifted to minimise its mean in-plane distance from the preceding ring, while preserving the contour direction. Corresponding points on adjacent rings are then joined by triangles, and the first and last rings are closed with planar caps. The resulting loft is rasterised on a 1.0 mm isotropic grid and extracted again with marching cubes so that its mesh density is comparable with the other evaluation meshes. It then passes through the same watertight repair and smoothing procedure used for the model output. The model and the lofting baseline are evaluated on the same patients against the same segmentation-derived geometry and with the same surface, overlap, and volume metrics. This comparison tests learned geometric completion against direct linear interpolation between the observed contours; it does not isolate the contribution of individual model components.

3.5. Wall-Thickness Evaluation Protocol

The wall-thickness evaluation is performed on the model-generated geometry. The segmentation volume is used to extract contour rings and to orient the AHA regional coordinate system, but the measured surfaces are the endocardial and epicardial meshes produced by the trained model. This distinction is important: the evaluation tests thickness estimation on the reconstructed implicit model, not directly on raw segmentation meshes. Myocardial wall thickness is a clinically relevant morphological measurement, but the present evaluation does not assess diagnostic performance for infarction, scarring, or hypertrophy. The evaluation is therefore organised in three steps: first the algorithms that measure thickness on the geometry, then the resulting local three-dimensional thickness field, and finally its aggregation into the clinical AHA-17 regional summary.

3.5.1. Selected thickness methods

Four wall-thickness algorithms are evaluated on the reconstructed geometry. They are chosen to span the main methodological families used to measure thickness: volumetric distance fields, ray intersections, and partial-differential-equation (PDE) formulations. The four are the Laplace field method, the Yezzi–Prince Eulerian PDE method, the signed-distance cone-ray method, and the Euclidean Distance Transform (EDT) boundary sum. No method is assumed in advance to be the best one. All four are computed on the same geometry and compared under the same conditions, and the one that proves most consistent is then used for the regional analysis reported in Chapter 4. Each method returns local thickness values in millimetres at endocardial vertices or projected endocardial locations. The outputs are summarised with mean, median, standard deviation, fifth percentile, ninety-fifth percentile, finite-value fraction, and runtime, and are visualised as 3D colour maps and AHA-17 bullseye plots.

Table 3.6. Wall-thickness methods selected for evaluation on the model-generated LV geometry, with the Euclidean Distance Transform boundary sum kept only as a baseline.

Method	Category	Main measurement principle
Laplace field	PDE	Transmural field lines from a harmonic potential
Yezzi–Prince	PDE	Eulerian transport equations along the transmural field
SDF cone rays	Ray / SDF	Median of multiple cone rays around the surface normal
EDT boundary sum	Volumetric	Sum of endocardial and epicardial distance transforms

The EDT boundary sum computes, at each myocardial voxel x, the sum of the distances to the endocardial and epicardial surfaces,

$$t^{\text{EDT}}(x) = D_{\text{endo}}(x) + D_{\text{epi}}(x), \tag{3.14}$$

where D_{endo} and D_{epi} are the Euclidean distance transforms of the two boundaries. This is exact for planar parallel walls but is limited by voxel resolution and slightly overestimates thickness in strongly curved regions.

The Laplace field method defines a smooth scalar field ψ over the myocardium by solving

$$\nabla^2\psi = 0, \qquad \psi|_{\text{endo}} = 0, \qquad \psi|_{\text{epi}} = 1. \tag{3.15}$$

The local wall thickness is then estimated from the gradient magnitude along the transmural field. This family of methods is attractive because harmonic field lines do not cross, but numerical errors can appear where the wall is thin or where the field gradient is small (Jones et al., 2000). The Yezzi–Prince approach addresses this by solving Eulerian transport equations along the same transmural field and combining the endocardial and epicardial travel times (Yezzi & Prince, 2003). In simplified form,

$$\nabla\psi \cdot \nabla u = -1, \qquad t = u + v, \tag{3.16}$$

where u and v are the propagation distances from opposite boundaries.

The cone-ray method uses the endocardial surface normal but reduces sensitivity to a single noisy ray. Inspired by the shape diameter function (Shapira et al., 2008), for each endocardial point several rays are cast inside a cone around the normal direction, and the median valid hit distance is used:

$$t_i^{\text{cone}} = \text{median}_{k=1}^{K}\, d_{ik}, \qquad K = 7. \tag{3.17}$$

In the notebook implementation the cone half-angle is $30°$. This makes the estimate more robust than a single normal ray while still preserving an approximately transmural interpretation.

Figure 3.8 contrasts how the four estimators measure thickness on the same wall cross-section. The EDT method sums the distances from an interior point to both boundaries; the Laplace method follows the transmural streamlines of a harmonic potential whose nested iso-surfaces are shown as dashed lines; the Yezzi–Prince method integrates two travel times u and v along that same field and adds them; and the cone-ray method casts a fan of rays around the endocardial normal and keeps the median hit distance.

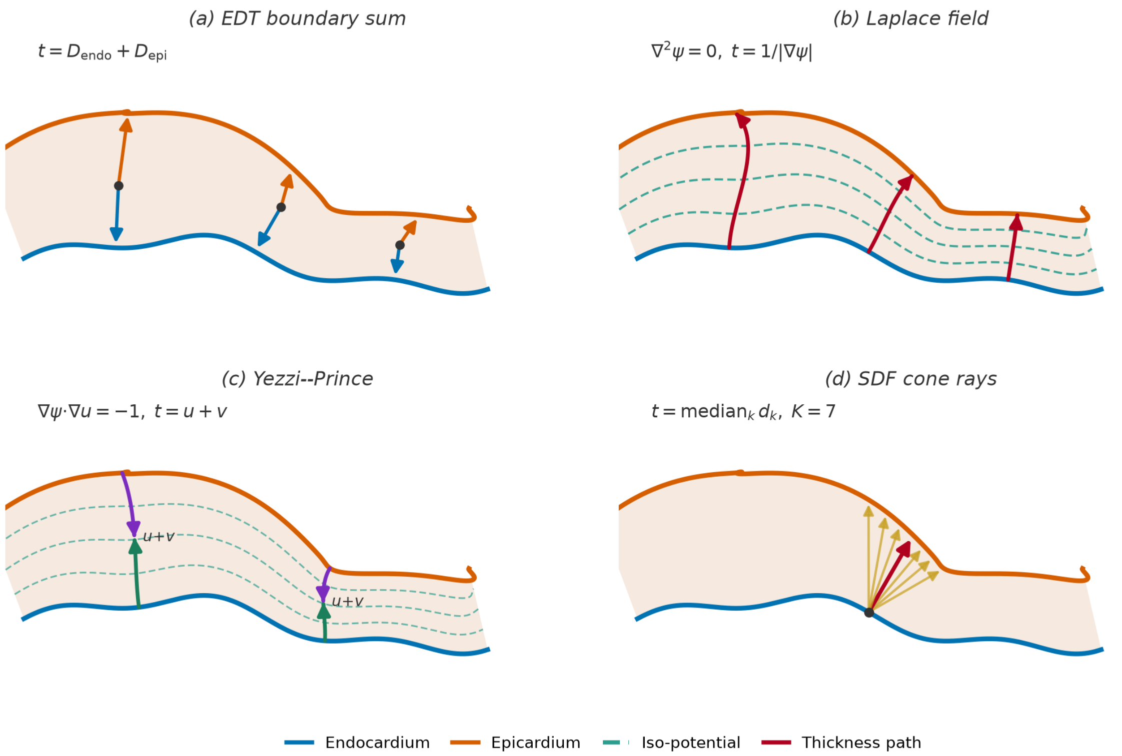

Figure 3.8. Conceptual comparison of the four wall-thickness estimators on a single myocardial cross-section, with the endocardium in blue and the epicardium in orange. (a) The EDT boundary sum adds the distances from an interior point to each boundary. (b) The Laplace field method follows the curved transmural streamlines of a harmonic potential, which cross the iso-potential surfaces (dashed lines) at right angles, and estimates thickness from the gradient magnitude. (c) The Yezzi–Prince method propagates two travel times u and v from opposite boundaries along that same curved field and sums them. (d) The SDF cone-ray method casts a fan of rays within a cone around the endocardial normal and takes the median intersection distance (highlighted), making it robust to any single ill-oriented ray.

3.5.2. Local three-dimensional wall thickness

Each of the selected methods produces, before any regional summary, a continuous thickness field defined over the whole reconstructed geometry. After the evaluation post-processing described in Section 3.4.4, the endocardial and epicardial surfaces are densely sampled and a thickness value is attached to each endocardial vertex. This produces a local three-dimensional wall-thickness map rather than a handful of per-slice numbers, as shown in Figure 3.9, where the LV surface is coloured by local thickness from the thin apical region to the thicker basal wall.

This local map is the primary output of the wall-thickness stage. Presenting the measurement first as a full surface field, and only afterwards as regional segment averages, makes the spatial distribution of thickness explicit: localised thinning or thickening remains visible on the surface even when it would be averaged away inside a single AHA segment. The colour map uses a fixed thickness range across cases so that maps from different patients and cardiac phases can be compared directly.

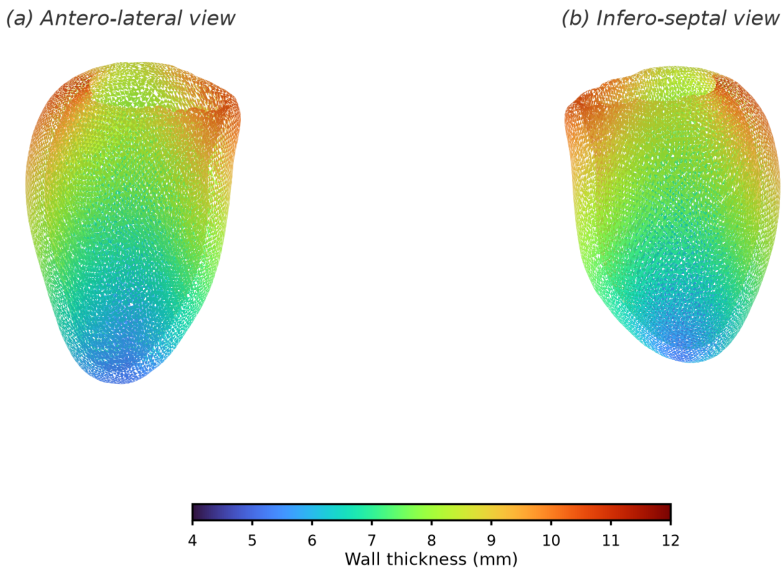

Figure 3.9. Local three-dimensional wall-thickness map on the reconstructed LV endocardial surface, shown from two opposing viewpoints. Each vertex is coloured by its local wall thickness on a fixed millimetre scale, so that the continuous spatial variation from the thin apex to the thicker base is visible directly on the geometry rather than only as regional averages.

3.5.3. Regional AHA-17 mapping

For the final regional analysis, the surface field from the method judged most consistent in the comparison is aggregated into the AHA 17-segment model, the accepted clinical language for reporting regional myocardial measurements (Cerqueira et al., 2002). Conventionally the wall thickness in each segment is read semi-manually from a few SAX slices, which is observer-dependent, coarsely sampled, and sensitive to the choice of slice and measurement point. Here the same segment values are produced objectively from the selected method's field of Section 3.5.2, by cutting the reconstructed surface into the AHA segments and averaging within each one.

The cut is defined in the LV's own anatomical frame, which first requires identifying the long-axis direction, distinguishing base from apex, and fixing a septal reference direction; the wall-thickness notebook includes an anatomical QA step that visualises this base/apex orientation and septal alignment before the segments are formed. The surface is then partitioned in two stages, as illustrated in Figure 3.10. First, along the long axis, the endocardial vertices are split by their normalised apex–base position into a basal ring, a mid-cavity ring, an apical ring, and a small apical cap. Second, within each ring the vertices are divided angularly around the long-axis centreline: the basal and mid rings are cut into six sectors each and the apical ring into four, while the apical cap forms the single seventeenth segment. Numbering the sectors from the anterior–septal reference and unrolling the rings onto concentric circles yields the familiar bullseye, in which the basal ring is the outer annulus, the mid ring the middle annulus, the apical ring the inner annulus, and the apical cap the central disc.

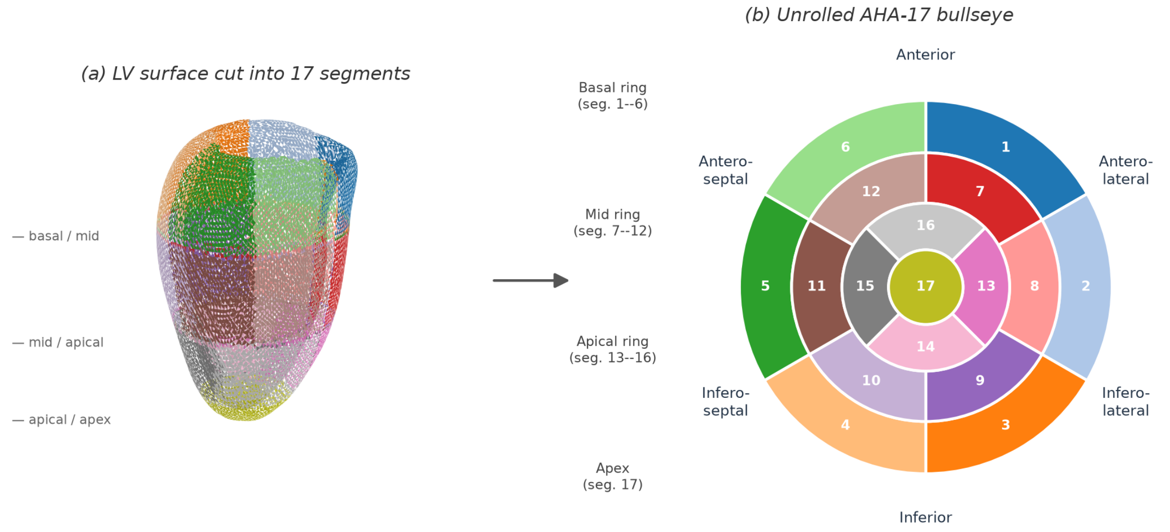

Figure 3.10. Cutting the reconstructed LV surface into the AHA-17 segments. Panel (a) shows the endocardial surface partitioned along the long axis into basal, mid, apical, and apical-cap levels and angularly into sectors, with each of the 17 regions drawn in a distinct colour. Panel (b) unrolls the same colour-coded regions onto the standard AHA-17 bullseye, where the basal ring (segments 1–6) forms the outer annulus, the mid ring (7–12) the middle annulus, the apical ring (13–16) the inner annulus, and the apical cap (17) the central disc; the anatomical wall regions are labelled around the disc.

The AHA aggregation does not replace the per-vertex thickness maps; it provides a clinically familiar regional summary of the same local measurements. For each method, the thickness values of the vertices falling in a segment are combined using robust summaries such as the mean and percentiles, so that local outliers do not dominate the regional interpretation.

This evaluation should be interpreted as an algorithmic comparison and reproducibility study rather than as clinical validation. Unless expert manual wall-thickness measurements and independent 3D targets are added, the thesis can conclude only that the model provides a locally resolved computational framework for reconstruction consistency; it should not claim anatomical truth, diagnostic utility, or replacement of clinical measurement protocols.

CHAPTER 4

Results and Evaluation

This chapter presents the experimental evaluation of the approach described in Chapter 3.

4.1. Experimental Setup

The evaluation uses the final model, obtained by pre-training on the synthetic SSM meshes and then fine-tuning on the real end-diastolic (ED) and end-systolic (ES) data, as described in Chapter 3. The datasets and patient-level splits are reported in Section 3.2.6. The results below are computed on a cohort of 30 ACDC patients, comprising 20 normal (NOR) and 10 hypertrophic cardiomyopathy (HCM) cases: each is reconstructed at end-diastole for the geometric metrics and at both phases for the wall-thickness and regional analyses. The HCM group is included so that the model can be examined on abnormal as well as on normal LV geometry. Reconstruction accuracy is measured against a segmentation-derived comparator obtained by marching-cubes extraction on the input segmentations, which is the only reference available for these scans. The same thickness estimators are applied to the model and derived geometries. Three questions guide the evaluation: how faithfully the model reconstructs the LV surfaces, whether it represents the ED–ES change while maintaining a valid wall, and how closely the resulting thickness measurements agree with those from the derived geometry.

4.1.1. Metrics

Reconstruction quality is measured with complementary surface and volume quantities, following the metric taxonomy of Taha and Hanbury (2015) and the conventions used for cardiac segmentation (Bernard et al., 2018) and for implicit-surface learning (Mescheder et al., 2019). Together they describe typical and tail surface error, volumetric overlap, and systematic size bias.

The *Chamfer distance* is the average distance between the two surfaces, taken in both directions. It is reported for the endocardium and the epicardium in millimetres, and smaller is better.

The *Dice similarity coefficient* (DSC) measures the solid rather than the surface. After voxelisation it is twice the shared volume divided by the sum of the two volumes. It is computed for the endocardial cavity and for the myocardium, lies between zero and one, and larger is better. A shape can score well on overlap and still be uniformly too small, so the *volume ratio* is reported as well. It is the reconstructed volume divided by the reference volume, and a value of one means there is no systematic error in size. It is the only reported quantity that directly measures such a bias.

The *watertight rate* is the fraction of reconstructions that close into a surface with no holes. It is stated in the text as a precondition rather than as a score, because a mesh that is not closed has no volume for the DSC and the volume ratio to use.

The average symmetric surface distance (ASSD) provides a conventional check on the mean surface error, while the 95th-percentile Hausdorff distance (HD95) describes the tail of the error distribution. Both are reported with the Chamfer distance. The intersection over union is omitted because it is a direct transformation of the DSC; the F-score depends on one selected tolerance, and normal consistency measures orientation rather than positional accuracy.

Wall thickness is summarised by its mean, its standard deviation, and its 5th and 95th percentiles across the myocardial surface, all in millimetres. The percentiles describe the thin apical and thick basal extremes without being distorted by isolated outliers. When the two cardiac phases are compared, the absolute thickening $\Delta_{\text{ES}-\text{ED}}$ and the relative thickening as a percentage of the end-diastolic value are also reported. These are the standard descriptors of regional contractile function.

The four wall-thickness methods are compared with one another on the reconstructed geometry, using the agreement framework of Bland and Altman (1986) instead of a comparison of mean values. The reported quantities are the *Pearson correlation coefficient* r, the *mean absolute error* (MAE) and *root-mean-square error* (RMSE) in millimetres, and the Bland–Altman *bias* with its *95% limits of agreement* (LoA) (bias $\pm 1.96\,\sigma$). Absolute consistency is summarised by the two-way *intraclass correlation coefficient* (ICC), read against the thresholds of Koo and Li (2016). These are reported together because correlation alone does not show agreement: two methods can rise and fall together and still differ by a constant amount, which only the bias reveals.

4.1.2. Comparators

No paired independent 3D mesh exists for the clinical scans, as discussed in Section 3.2.1. The segmentation-derived geometry is therefore used as the reconstruction comparator. A matched linear contour-lofting baseline is also evaluated. It uses the same input contour rings as the model and the same mesh repair and metric pipeline, as described in Section 3.4.4. Wall thickness is measured on both geometries with the four algorithms selected in Section 3.5.1: the Laplace field method, the Yezzi–Prince method, the SDF cone-ray method, and the Euclidean Distance Transform (EDT) boundary sum. The thickness computed directly on the input label mask is reported as an anchor for the overall magnitude, as the boundary sum of the two distance transforms over every myocardial voxel.

That anchor is not a ground truth, because the input is sparse. Across the cohort only 8.6 slices on average cross the myocardium, from 6 to 14 depending on the patient. The slices are 10 mm apart in 29 of the 30 cases and 5 mm apart in the remaining one, while the in-plane pixel size is between 1.37 and 1.88 mm. Resampling such a stack to a 1.0 mm isotropic grid interpolates about nine voxels in ten along the long axis, so the anchor carries almost no measured information in that direction. Absolute millimetre values should therefore be read with care throughout this chapter.

4.2. Training

The final model is produced in two stages. A first stage pre-trains the signed-distance decoder for roughly 200 epochs on the synthetic SSM meshes. This stage provides a geometric prior, and its checkpoint initializes the second stage. The reported model is then fine-tuned on the real end-diastolic and end-systolic data described in Chapter 3. The available training history records the fine-tuning stage; no numerical claim is made about the pre-training loss because its per-epoch log was not retained.

Fine-tuning runs for 776 epochs with the AdamW optimiser (learning rate 2×10^{-5}, weight decay 5×10^{-4}) under a cosine-annealing schedule with warm restarts, mixed-precision arithmetic, and an effective batch size of 32. The composite objective is the five-term loss of Equation (3.12), combining surface, dense SDF, Eikonal, off-surface, and wall-thickness supervision. Figure 4.1 plots the total loss and its principal components together with two optimisation diagnostics. The total training loss decreases from 10.07 at the first fine-tuning epoch to 0.91, and the validation loss settles at 0.70, with the best checkpoint selected at epoch 695 before mild over-fitting appears. The surface and Eikonal terms fall by more than an order of magnitude, indicating successful optimisation of the surface-fitting and Eikonal objectives. Consistently, the mean gradient norm $\|\nabla f\|$ stabilises at 0.996, close to the unit value encouraged by the Eikonal term.

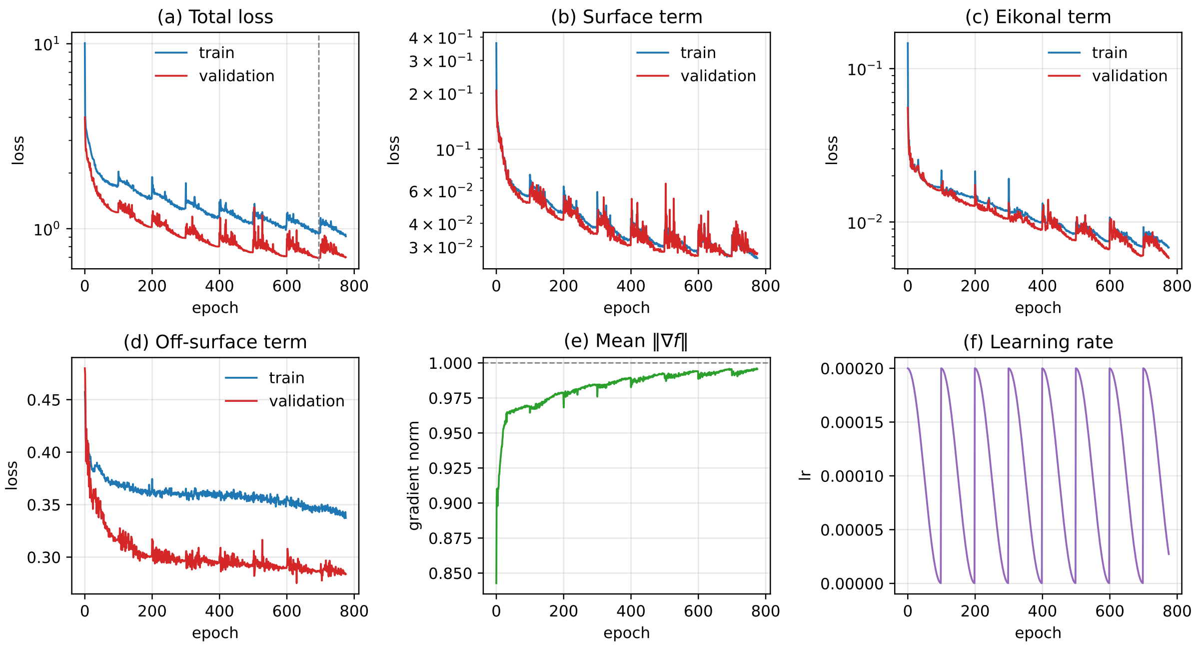

Figure 4.1. Fine-tuning history. Panels (a)–(d) show the total loss and its surface, Eikonal, and off-surface components for the training and validation splits; the dashed vertical line marks the best epoch. Panels (e)–(f) show the mean field gradient norm, whose target is one, and the cosine-annealing learning-rate schedule. The synthetic pre-training stage is not shown because its optimisation history was not retained.

4.3. Results

This section reports first the geometric quality of the reconstruction, then the wall-thickness measurements obtained from the reconstructed geometry, and finally the regional analysis over the AHA 17-segment model.

4.3.1. Reconstruction Quality

Table 4.1 summarises reconstruction accuracy over the evaluation cohort. Each case is reconstructed at end-diastole from its SAX contours and compared with the reference geometry. The reported metrics describe typical and tail surface error, volumetric overlap, and systematic volume bias.

The Chamfer distance gives the error at a typical point. It is 1.22 ± 0.21 mm on the endocardium and 1.05 ± 0.18 mm on the epicardium, so an average point of the reconstructed surface lies about one millimetre from the reference. This is finer than the in-plane pixel size of the source images, which is between 1.37 and 1.88 mm.

The ASSD is similar, at 1.21 ± 0.23 mm and 1.09 ± 0.20 mm for the two surfaces. The HD95 is larger, at 3.30 ± 0.91 mm on the endocardium and 3.47 ± 1.41 mm on the epicardium. Most of each surface is therefore close to the comparator, while a smaller set of locations carries errors of several millimetres.

The Dice score gives the share of the enclosed volume that is recovered. It is 0.93 ± 0.02 for the cavity and 0.85 ± 0.02 for the myocardium. The myocardial value is lower because the myocardium is a thin shell. A boundary shift of one millimetre removes a much larger fraction of a wall a few millimetres thick than of a compact cavity, so the two numbers are not directly comparable and the gap between them is expected.

The volume ratio shows whether the reconstruction is systematically the wrong size. It is 0.92 ± 0.03 for the cavity and 1.00 ± 0.03 for the epicardial solid. The epicardium is therefore the right size on average, while the cavity is consistently about 8% too small. The Chamfer distance cannot show this, because it would report the same sub-millimetre error for a uniform inward shift as for random noise of the same size.

Every endocardial and epicardial reconstruction in the cohort also closes into a manifold, a watertight rate of 100% on both surfaces after the repair stage. This makes the enclosed volumes well defined and therefore makes the volume ratios meaningful.

Splitting the cohort by group shows that the reconstruction degrades only mildly on the abnormal ventricles. The endocardial Chamfer distance is 1.12 mm on the normal cases and 1.43 mm on the hypertrophic ones, and the endocardial Dice falls from 0.94 to 0.91. The myocardial Dice is slightly higher on the HCM group (0.87 against 0.84), because a thicker wall is easier to overlap. The remaining differences are consistent with the more irregular wall geometry of the hypertrophic cases.

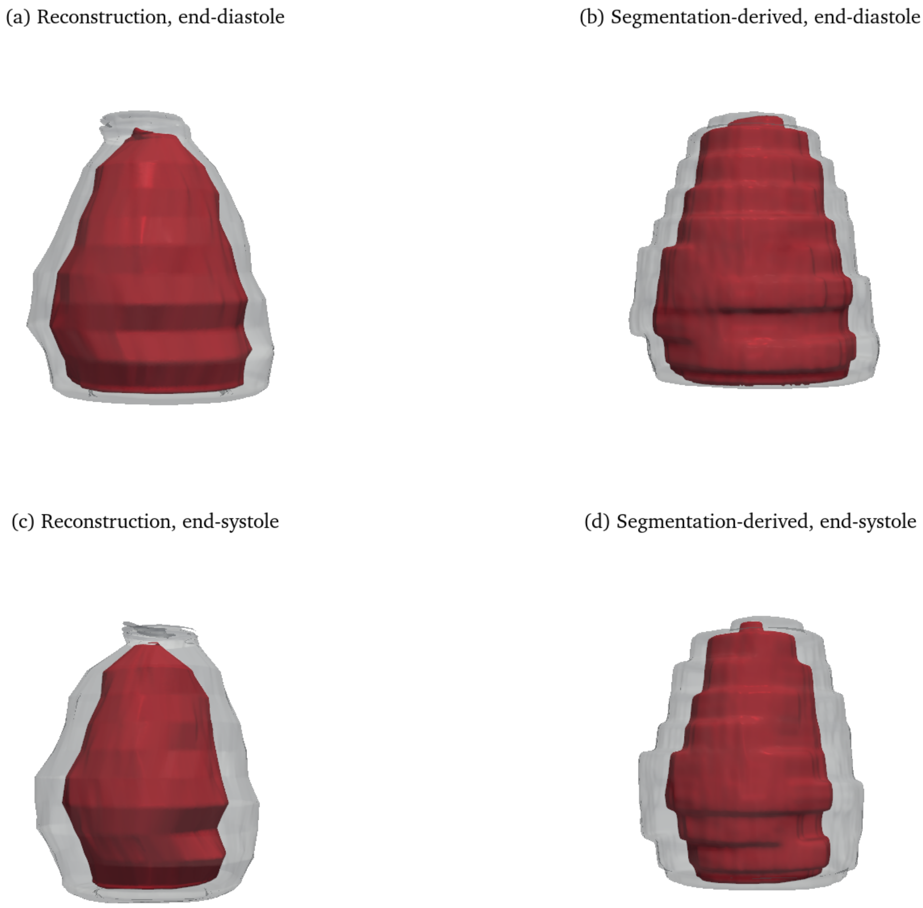

Figure 4.2. Reconstructed and segmentation-derived left-ventricular surfaces at both cardiac phases, for one normal case of the evaluation cohort. Panels (a) and (c) show the reconstruction, panels (b) and (d) the geometry derived from the same segmentation. The opaque red surface is the endocardium and the translucent grey shell is the epicardium; all eight surfaces are watertight, and the four panels share one scale and viewpoint, with the base upwards. The end-systolic cavity is smaller and the wall thicker, consistent with contraction. The fine circumferential banding on the reconstruction marks the through-plane spacing of the input SAX slices, between which the field is interpolated; the segmentation-derived surfaces instead show the voxel staircase of the label mask.

Figure 4.2 shows the reconstructed endocardial and epicardial surfaces for one normal case at both cardiac phases, beside the geometry derived from the same segmentation. The end-systolic cavity is visibly smaller than the end-diastolic one, and the surrounding wall correspondingly thicker. This shows that the pipeline reconstructs the expected phase-dependent change. Both reconstructed surfaces are zero-level sets of the same continuous field rather than stacks of per-slice contours, and the comparison with the segmentation-derived surfaces shows that the reconstruction recovers the same overall shape while smoothing the voxel staircase of the label mask.

Table 4.1. Reconstruction quality of the final model over the 30-patient evaluation cohort at end-diastole, reported as mean $\pm$ standard deviation across patients. The metrics report typical and tail surface error, volumetric overlap, and systematic volume bias. Distances are in millimetres; the Dice similarity coefficient and the volume ratio are dimensionless. Every reconstruction in the cohort is watertight on both surfaces.

Metric	Endocardium	Epicardium
Chamfer distance (mm)	1.22 ± 0.21	1.05 ± 0.18
ASSD (mm)	1.21 ± 0.23	1.09 ± 0.20
HD95 (mm)	3.30 ± 0.91	3.47 ± 1.41
Volume ratio	0.92 ± 0.03	1.00 ± 0.03
	Cavity	Myocardium
Dice similarity coefficient	0.93 ± 0.02	0.85 ± 0.02

4.3.1.1. *Matched contour-lofting baseline* Table 4.2 compares the model with direct linear lofting between the same input contours. Both methods are evaluated on all 30 patients against the same segmentation-derived geometry. Both also produce watertight endocardial and epicardial surfaces in every case after the shared repair stage.

Table 4.2. Reconstruction quality of the model and the matched linear contour-lofting baseline at end-diastole. Values are mean $\pm$ standard deviation across the 30 patients. The final column is the paired lofting-minus-model difference with a 95% percentile-bootstrap confidence interval from 10 000 patient-level resamples. Distances are in millimetres; Dice and volume ratio are dimensionless.

Metric	Structure	Model	Lofting	Difference (95% CI)
Chamfer distance	Endocardium	1.22 ± 0.21	1.60 ± 0.20	$+0.38$ $[0.28,\ 0.47]$
	Epicardium	1.05 ± 0.18	1.86 ± 0.22	$+0.81$ $[0.73,\ 0.89]$
ASSD	Endocardium	1.21 ± 0.23	1.55 ± 0.20	$+0.34$ $[0.23,\ 0.44]$
	Epicardium	1.09 ± 0.20	1.85 ± 0.24	$+0.77$ $[0.68,\ 0.85]$
HD95	Endocardium	3.30 ± 0.91	6.88 ± 1.91	$+3.59$ $[3.03,\ 4.17]$
	Epicardium	3.47 ± 1.41	8.35 ± 2.50	$+4.88$ $[4.18,\ 5.56]$
Dice	Cavity	0.93 ± 0.02	0.92 ± 0.01	-0.010 $[-0.016,\ -0.004]$
	Myocardium	0.85 ± 0.02	0.81 ± 0.04	-0.043 $[-0.054,\ -0.032]$
Volume ratio	Cavity	0.92 ± 0.03	0.89 ± 0.02	-0.034 $[-0.044,\ -0.024]$
	Epicardial solid	1.00 ± 0.03	0.89 ± 0.02	-0.115 $[-0.126,\ -0.105]$

The model has lower mean and tail surface error for both surfaces. The largest difference is in HD95: contour lofting reaches 6.88 mm on the endocardium and 8.35 mm on the epicardium, compared with 3.30 and 3.47 mm for the model. The paired confidence intervals exclude zero for every reported difference. The cavity Dice advantage is small, at 0.01, but the myocardial Dice improves from 0.81 to 0.85. Linear lofting also produces a volume ratio of about 0.89 for both solids, whereas the model reaches 1.00 for the epicardial solid. These results show that the learned reconstruction does more than direct linear interpolation and smoothing under this matched evaluation. In particular, it better controls between-slice surface excursions and epicardial volume. The comparison does not show which architectural component causes the improvement, and it does not establish superiority over more advanced reconstruction methods.

4.3.1.2. *Patient-level error analysis* The patient-level distribution clarifies where the averages are reliable and where they hide difficult cases. Endocardial Chamfer distance ranges from 0.88 to 1.77 mm and epicardial Chamfer distance from 0.76 to 1.43 mm. The corresponding maximum HD95 values are 6.22 and 6.95 mm, whereas myocardial Dice remains between 0.80 and 0.89. In particular, the case with the largest endocardial HD95 still reaches a myocardial Dice of 0.89, showing that a local surface excursion can coexist with good overall overlap. All 30 cavity volume ratios are below one, so the cavity shrinkage is systematic rather than being caused by a few outliers. The cached metrics do not identify the anatomical location of each surface error; attributing these tails specifically to the apex or base would require point-wise anatomical error maps.

Figure 4.3 shows the patient-level distributions behind these summaries and separates the normal and HCM groups without treating group membership as a reconstruction comparator.

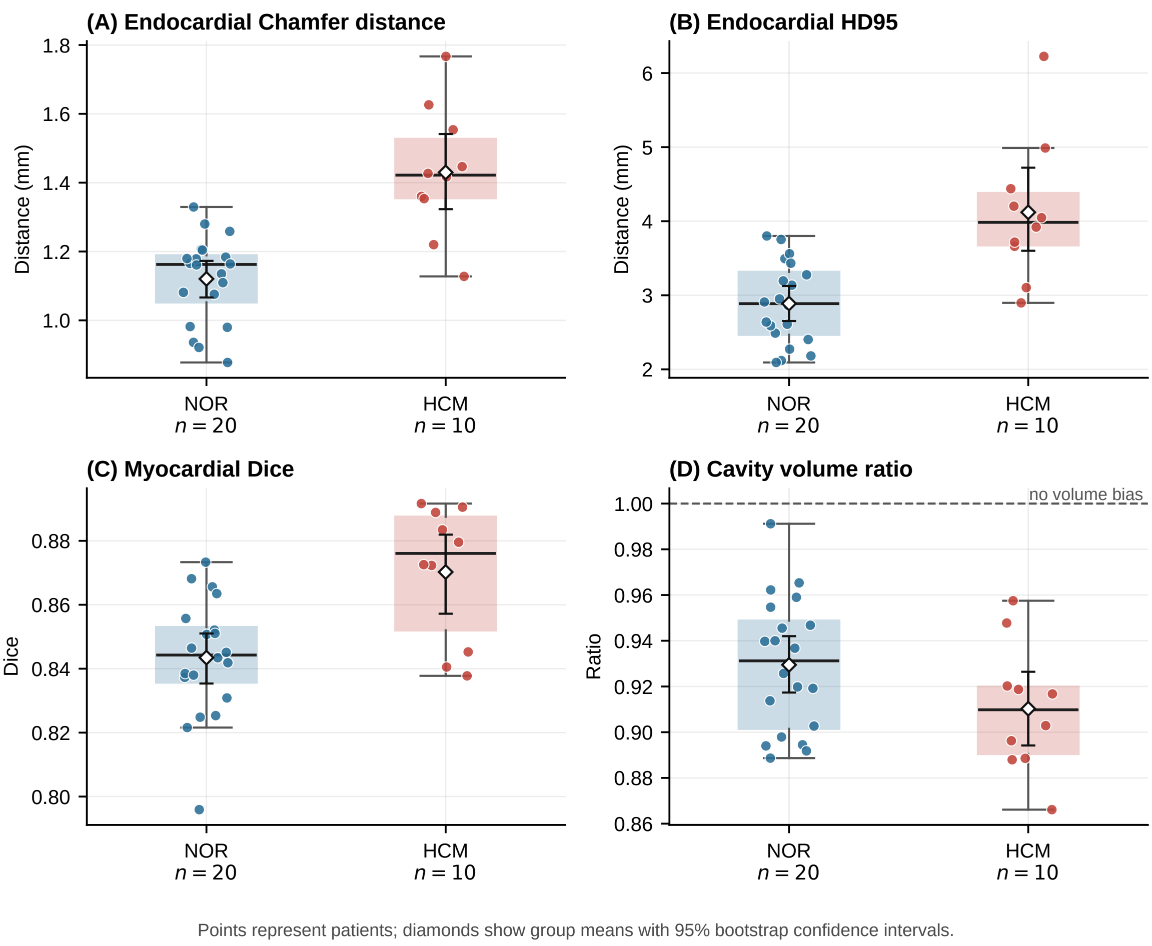

Figure 4.3. Patient-level end-diastolic reconstruction metrics for the normal and HCM groups. Points represent patients; diamonds and bars show group means with 95% bootstrap confidence intervals.

4.3.2. Wall-Thickness Measurement

The four selected methods are applied to both the reconstructed geometry and the voxel-based geometry derived from the same input segmentation of every patient at end-diastole. Table 4.3 reports, for each method and geometry, the cohort average of the per-patient mean, standard deviation, and 5th and 95th percentiles. The voxel measurements are a segmentation-derived comparator, not independent anatomical ground truth.

The three non-ray methods agree on the magnitude of the wall on both geometries. On the reconstructed geometry, the Laplace field reads 9.18 mm, the EDT boundary sum 9.55 mm, and Yezzi–Prince 10.01 mm. The corresponding voxel-based values are 8.60, 8.63, and 9.14 mm. The SDF cone-ray method stands apart on both geometries, reading 12.47 mm on the reconstruction and 9.83 mm on the voxel-based geometry.

Dispersion separates the methods the same way. The per-patient standard deviation is 1.64 mm for Yezzi–Prince, 1.66 mm for the EDT boundary sum and 2.14 mm for the Laplace field, but 4.87 mm for the cone-ray estimator. Its 5th-to-95th-percentile span, 8.01 to 20.68 mm, is about twice as wide as that of any other method. The ray-based estimator is therefore both the highest and the noisiest of the four.

Table 4.3. Wall-thickness statistics at end-diastole over the 30-patient evaluation cohort. Each entry is the cohort average of the corresponding per-patient statistic, in millimetres, before any calibration. "Voxel" denotes measurements on geometry derived from the input segmentation; it is included as a comparator, not as independent anatomical ground truth.

Method	Geometry	Mean	Std.	p_5	p_{95}
Laplace field	Model	9.18	2.14	4.75	12.20
	Voxel	8.60	2.18	5.22	12.42
Yezzi–Prince	Model	10.01	1.64	7.41	12.74
	Voxel	9.14	2.07	5.98	12.82
SDF cone rays	Model	12.47	4.87	8.01	20.68
	Voxel	9.83	2.43	6.44	13.92
EDT boundary sum	Model	9.55	1.66	6.62	12.17
	Voxel	8.63	2.01	5.33	12.08

Applying the same estimator to the two geometries gives a direct measure of reconstruction consistency. The reconstructed geometry has a higher mean thickness for every method: $+0.58$ mm for the Laplace field, $+0.87$ mm for Yezzi–Prince, $+2.65$ mm for SDF cone rays, and $+0.91$ mm for the EDT boundary sum. Despite these positive offsets, the per-patient mean values are strongly correlated between the model and voxel-based geometries ($r = 0.99$, 0.99, 0.95, and 0.99, respectively). Thus, the reconstruction broadly preserves between-patient variation while shifting the absolute thickness scale; this is agreement with a segmentation-derived comparator, not validation against true anatomy.

Table 4.4 compares the three other methods with the Laplace field point by point over the myocardial surface of all 30 patients. The two remaining non-ray methods track it closely. The EDT boundary sum differs from it by 0.81 mm on average with a bias of $+0.37$ mm, and Yezzi–Prince by 0.89 mm with a bias of $+0.83$ mm; both reach ICC $= 0.77$. The cone-ray method differs by 3.38 mm on average, with a bias of $+3.31$ mm, limits of agreement of ± 12.74 mm and ICC $= 0.28$. Read against the thresholds of Koo and Li (2016), the first two show moderate to good agreement with the Laplace field and the third poor agreement. The correlations make the same point in a different way: all three exceed $r = 0.44$, but r alone would not have revealed a bias of more than three millimetres.

The Laplace field is the method used for the regional analysis of Section 4.3.3. The reason is methodological rather than numerical. It is the established reference for transmural thickness in cardiac imaging (Jones et al., 2000), and it measures thickness along the streamlines of a harmonic field spanning the two surfaces, so the path stays inside the wall wherever the wall curves. A straight ray does not: it leaves the myocardium early or late in curved regions, which is what the cone-ray numbers above show. The results here are consistent with that choice, since the Laplace field sits with the two non-ray methods rather than with the outlier. No claim is made that it is closer to true anatomy, because no independent measurement of the anatomy exists for these patients.

The decoder returns a continuous wall-offset field rather than a geometric thickness measurement. Figure 4.4 visualises this field for the representative case at both cardiac phases, in the same form as the methodology illustration of Figure 3.9 but computed on the model output instead of on the statistical-shape-model mean mesh. The decoder offset is defined only up to the normalisation applied to the input contours, so its absolute level is calibrated per phase to the mean of the distance-transform reference measured on the input segmentation, exactly as the cohort pipeline does; the calibration factor is close to two for both phases. The figure is therefore a calibrated analytic offset map, not a wall-thickness measurement or an independently predicted absolute magnitude. The residual circumferential banding visible on the surfaces marks the through-plane spacing of the input SAX slices, where the field is interpolated between observations.

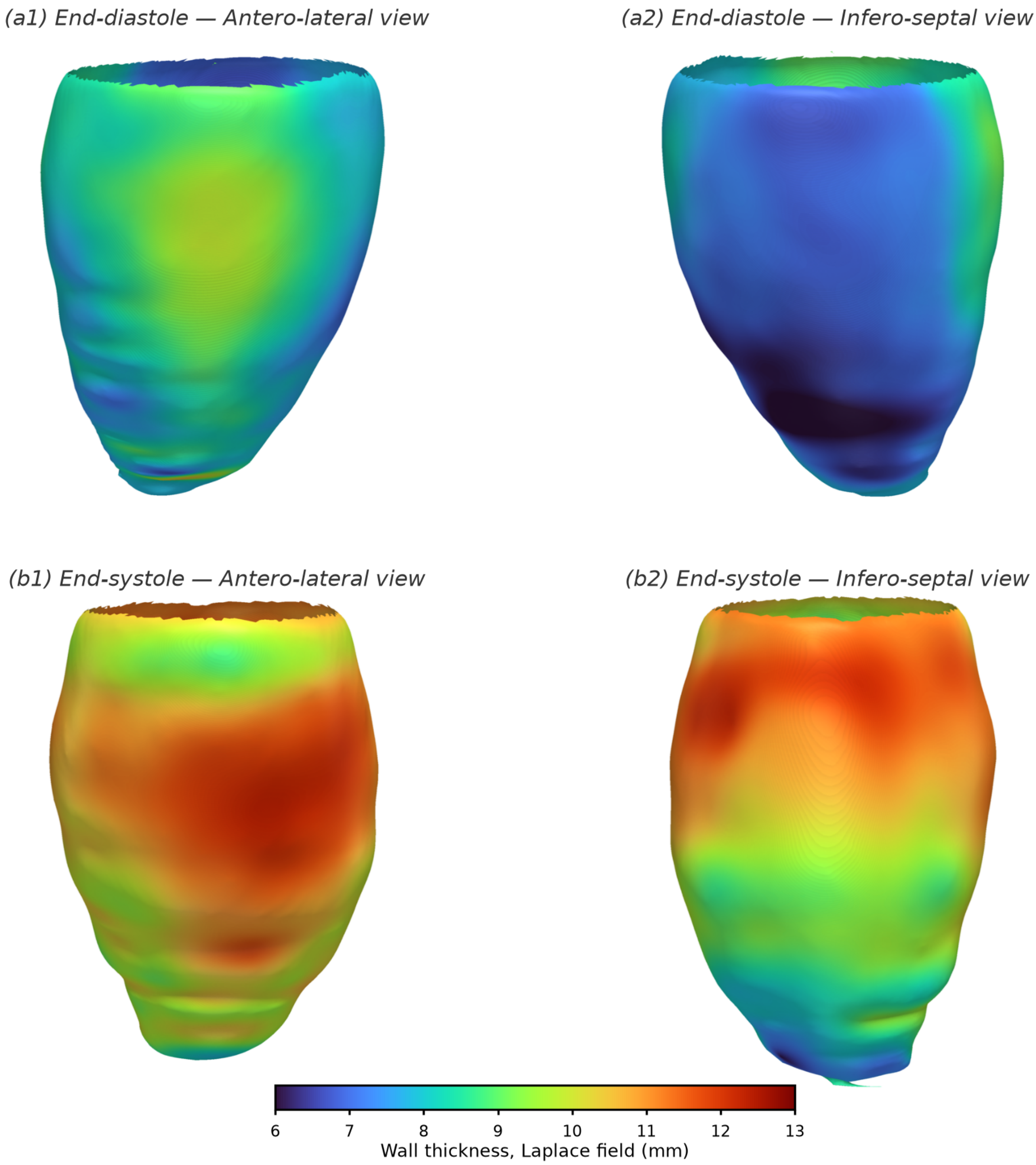

Figure 4.4. Calibrated analytic offset map on the reconstructed endocardial surface of one representative case from the ACDC cohort, at end-diastole (top) and end-systole (bottom), each seen from two opposing viewpoints. Every vertex is coloured by the decoder wall offset evaluated at that point, calibrated per phase to the distance-transform reference of the input segmentation, on a scale shared by all four panels. The absolute colour level is therefore inherited from that reference. The surface is trimmed at the most basal input contour, so the base ends in an open rim and only the observed part of the wall is shown; above that plane the field is pure extrapolation.

Figure 4.4 and Table 4.7 intentionally report different quantities. The figure visualises the calibrated decoder offset to show the spatial pattern emitted directly by the model. The table reports Laplace-derived wall thickness measured between the post-processed endocardial and epicardial meshes. The offset map is therefore a qualitative model-output visualisation; it does not provide the values used for the AHA-17 analysis.

Table 4.4. Agreement of each wall-thickness method with the Laplace field, over the myocardial surface points of the 30 patients at end-diastole on the reconstructed geometry. MAE, RMSE, bias and the 95% LoA are in millimetres; the Pearson correlation r and the intraclass correlation coefficient (ICC) are dimensionless. The Laplace field is the common comparator, not a ground truth.

Method	r	MAE	RMSE	Bias (95% LoA)	ICC
Yezzi–Prince	0.80	0.89	2.14	$+0.83$ (± 3.86)	0.77
SDF cone rays	0.44	3.38	7.29	$+3.31$ (± 12.74)	0.28
EDT boundary sum	0.78	0.81	2.07	$+0.37$ (± 3.99)	0.77

Table 4.5 extends the same comparison to ES. The broad ordering is preserved, but the separation between methods grows. At ES the Laplace mean differs from the voxel comparator by -0.23 mm, while the offsets are $+1.42$ mm for Yezzi–Prince, $+1.01$ mm for the EDT boundary sum, and $+5.35$ mm for cone rays. The cone-ray discrepancy therefore increases from 2.65 mm at ED to 5.35 mm at ES, indicating greater sensitivity to the more contracted systolic geometry.

Table 4.5. Cohort average of the per-patient mean wall thickness at ES. Bias is model minus derived geometry. All values are in millimetres; the derived geometry is a segmentation-based comparator, not an independent reference.

Method	Model	Derived	Bias
Laplace field	13.73	13.96	-0.23
Yezzi–Prince	15.86	14.44	$+1.42$
SDF cone rays	21.41	16.05	$+5.35$
EDT boundary sum	14.21	13.21	$+1.01$

The finite-value fraction is nearly identical across the four methods within each phase and geometry: approximately 83.7% at ED and 89.3% at ES on the reconstructed geometry, and 76.6% and 78.4% on the voxel geometry. It therefore describes the common evaluated surface support and does not distinguish method reliability in this cohort. The magnitude and agreement results, rather than this fraction, motivate the use of Laplace for the regional analysis.

Figure 4.5 shows the paired patient values behind the phase means. Laplace remains close to the identity line at both phases, whereas cone rays show the largest positive displacement, especially at ES.

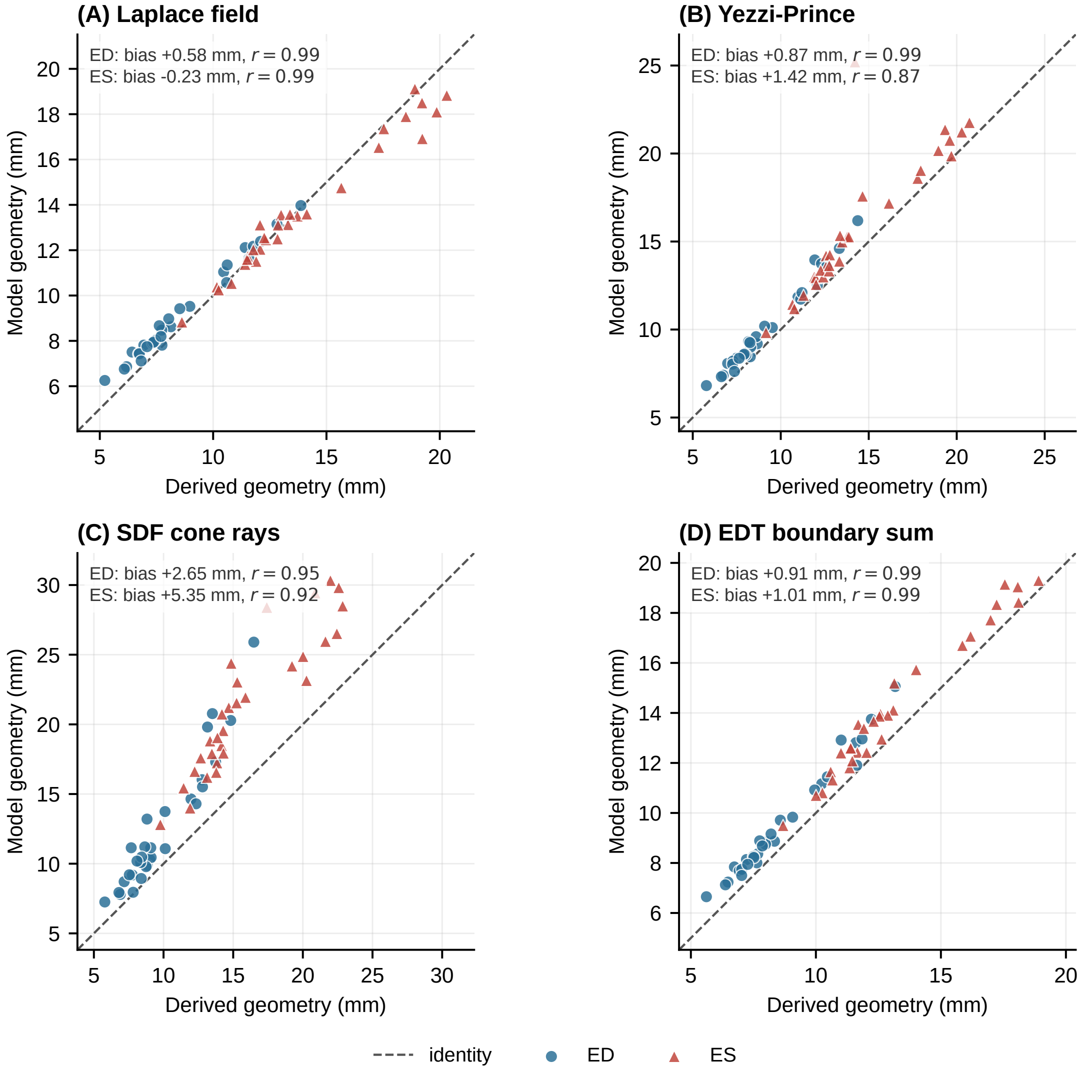

Figure 4.5. Patient-level mean wall thickness on model and segmentation-derived geometry at ED and ES. The dashed identity line denotes equal values; annotations report model-minus-comparator bias and Pearson correlation.

4.3.3. Regional Analysis over the AHA 17-Segment Model

To localise the measurements anatomically, the reconstructed surface is partitioned into the seventeen segments of the American Heart Association (AHA) model, assigning each myocardial vertex to a basal, mid-cavity, apical, or apical-cap segment from its normalised long-axis height and its circumferential angle relative to the right-ventricular insertion.

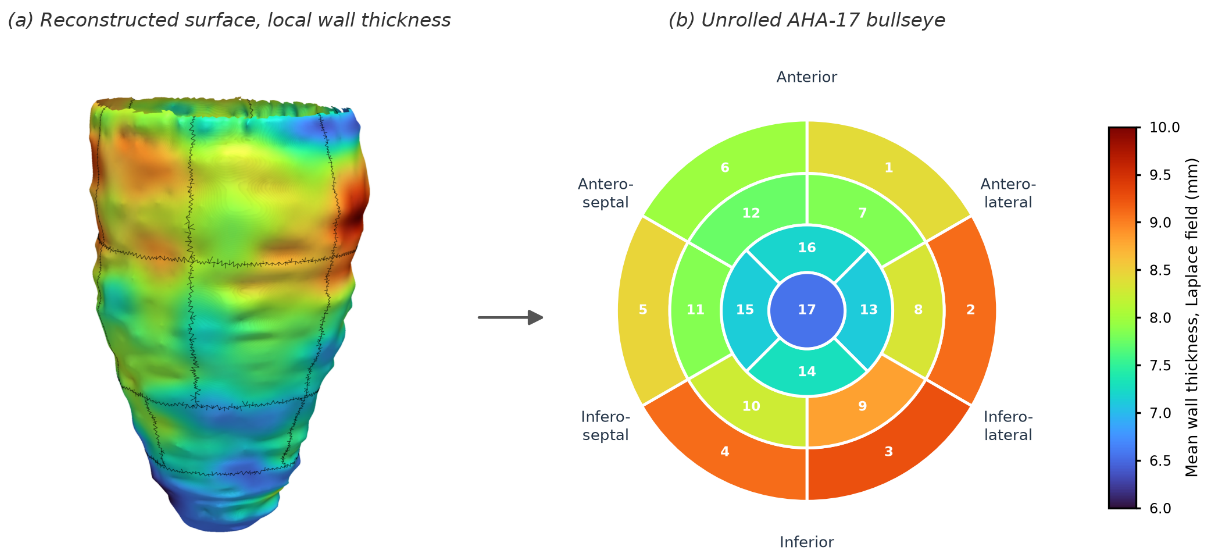

Figure 4.6. AHA 17-segment analysis of the reconstructed left-ventricular geometry at end-diastole for the representative case. Panel (a) shows the reconstructed surface partitioned into the seventeen segments, each coloured by its mean wall thickness measured with the Laplace field method and separated from its neighbours by a black boundary line; panel (b) unrolls the same segment values onto the standard bullseye, where basal segments form the outer ring (1–6), mid-cavity segments the middle ring (7–12), apical segments the inner ring (13–16), and the apical cap the centre (17). The two panels share the colour scale, so the bullseye is a flattening of the geometry beside it rather than an independent summary.

Figure 4.6 shows the partition applied to the reconstructed geometry of the representative case together with a bullseye of the Laplace-derived wall thickness. The regional results are then read in two steps.

Table 4.6 first compares the reconstruction with the voxel-based geometry over the whole cohort, which shows how far the regional pattern depends on the reconstruction at all. Table 4.7 then reports the reconstruction alone, split into the normal and the hypertrophic patients, which shows whether the model reacts to an abnormal ventricle or simply reproduces an average one. All values are measured with the Laplace field, for the reasons given in Section 4.3.2.

Table 4.6. Regional wall thickness over the AHA 17-segment model for the 30-patient cohort, comparing the reconstructed model geometry with the voxel-based geometry derived from the same input segmentation. Both are measured with the Laplace field. Columns give the mean end-diastolic (ED) and end-systolic (ES) thickness in millimetres and the relative thickening Δ as a percentage of the ED value. The voxel geometry is a segmentation-derived comparator, not an independent reference.

		Model geometry			Voxel geometry		
#	Segment	ED	ES	Δ (%)	ED	ES	Δ (%)
1	Basal Anterior	9.99	13.71	$+37.1$	10.05	14.22	$+41.4$
2	Basal Anteroseptal	10.15	13.40	$+32.1$	9.81	13.13	$+33.9$
3	Basal Inferoseptal	9.89	13.43	$+35.7$	9.72	13.63	$+40.2$
4	Basal Inferior	9.09	13.81	$+51.9$	8.51	13.77	$+61.9$
5	Basal Inferolateral	8.93	14.27	$+59.8$	8.45	14.25	$+68.6$
6	Basal Anterolateral	9.53	13.87	$+45.5$	9.07	14.23	$+56.9$
7	Mid Anterior	9.19	13.79	$+50.0$	8.21	14.01	$+70.6$
8	Mid Anteroseptal	10.37	14.68	$+41.6$	10.11	15.44	$+52.7$
9	Mid Inferoseptal	10.19	15.05	$+47.6$	10.16	15.30	$+50.6$
10	Mid Inferior	9.25	14.41	$+55.8$	8.48	14.35	$+69.3$
11	Mid Inferolateral	9.09	14.96	$+64.6$	8.27	15.00	$+81.3$
12	Mid Anterolateral	8.99	14.17	$+57.6$	8.30	14.73	$+77.5$
13	Apical Anterior	8.50	13.63	$+60.3$	7.72	13.67	$+77.0$
14	Apical Septal	8.77	13.86	$+58.1$	8.23	13.86	$+68.4$
15	Apical Inferior	8.44	13.29	$+57.5$	7.55	13.24	$+75.3$
16	Apical Lateral	8.43	13.78	$+63.6$	7.58	14.21	$+87.5$
17	Apex	7.76	11.69	$+50.7$	6.98	12.64	$+81.1$
Overall		9.21	13.87	$+50.6$	8.66	14.10	$+62.8$

Over the whole cohort the two geometries agree on the size and the arrangement of the wall at end-diastole. The reconstruction averages 9.21 mm and the voxel-based geometry 8.66 mm, and both place the thickest wall in the septal mid-cavity segments and the thinnest at the apical cap, which reads 7.76 mm on the reconstruction and 6.98 mm on the voxel geometry. The reconstruction is the thicker of the two in sixteen of the seventeen segments, the single exception being the basal anterior segment.

The end-systolic phase is where the two geometries part. The voxel-based geometry thickens by 62.8% between the phases and the reconstruction by 50.6%, and the gap is widest at the apex, where the two report 81.1% and 50.7%. The reconstruction therefore describes the end-diastolic wall more faithfully than it describes the change between the phases, which is consistent with the end-systolic limitation discussed in Section 4.4. Both geometries are built from the same sparse SAX contours, so the voxel column is a segmentation-derived comparator and not an independent measurement of the anatomy.

Table 4.7. Regional wall thickness over the AHA 17-segment model for the reconstructed model geometry, measured with the Laplace field and reported separately for the 20 normal (NOR) and the 10 hypertrophic (HCM) patients. Each group reports mean end-diastolic (ED) and end-systolic (ES) thickness in millimetres, together with relative thickening Δ. The values derive from the same sparse SAX contours as the reconstruction, so they summarise the model output and are not an independent reference.

		Normal (NOR, $n = 20$)			Hypertrophic (HCM, $n = 10$)		
#	Segment	ED	ES	Δ (%)	ED	ES	Δ (%)
1	Basal Anterior	8.72	12.25	$+40.4$	12.54	16.62	$+32.6$
2	Basal Anteroseptal	8.81	12.10	$+37.4$	12.83	16.02	$+24.9$
3	Basal Inferoseptal	8.68	11.96	$+37.8$	12.32	16.35	$+32.8$
4	Basal Inferior	8.16	12.53	$+53.6$	10.96	16.37	$+49.4$
5	Basal Inferolateral	7.94	12.81	$+61.3$	10.91	17.19	$+57.5$
6	Basal Anterolateral	8.38	12.12	$+44.6$	11.83	17.37	$+46.9$
7	Mid Anterior	7.94	12.10	$+52.5$	11.70	17.16	$+46.6$
8	Mid Anteroseptal	8.55	12.98	$+51.7$	13.99	18.10	$+29.4$
9	Mid Inferoseptal	8.57	12.98	$+51.4$	13.43	19.17	$+42.8$
10	Mid Inferior	7.93	12.46	$+57.1$	11.89	18.32	$+54.1$
11	Mid Inferolateral	7.96	13.02	$+63.5$	11.35	18.84	$+66.0$
12	Mid Anterolateral	7.92	12.39	$+56.4$	11.13	17.73	$+59.3$
13	Apical Anterior	7.28	11.75	$+61.4$	10.96	17.40	$+58.7$
14	Apical Septal	7.49	12.00	$+60.2$	11.33	17.59	$+55.3$
15	Apical Inferior	7.28	11.72	$+61.1$	10.76	16.43	$+52.6$
16	Apical Lateral	7.33	12.03	$+64.0$	10.61	17.28	$+62.9$
17	Apex	6.80	10.32	$+51.8$	9.67	14.71	$+52.1$
Overall		7.99	12.21	$+52.9$	11.68	17.23	$+47.5$

The regional pattern is internally coherent in both groups. In the normal cases the end-diastolic thickness varies little around the base and the mid-cavity, from 7.92 to 8.81 mm, and tapers towards the apex, where the apical cap at 6.80 mm is the thinnest region. Every segment thickens from end-diastole to end-systole, and the thickening is graded along the long axis: the apical segments thicken by 52% to 64%, the mid-cavity segments by 51% to 64% and the basal segments by 37% to 61%. Over the normal group the mean thickness rises from 7.99 mm to 12.21 mm, a thickening of 52.9%.

The hypertrophic group is thicker in every segment. The largest differences are in the mid-cavity anteroseptal and inferoseptal segments, where the HCM group reaches 13.99 and 13.43 mm compared with 8.55 and 8.57 mm in the normal group. Table 4.8 evaluates this separation at the patient level rather than treating the 17 segments as independent observations.

Table 4.8. Patient-level contrasts between the normal and HCM groups. Values are mean $\pm$ standard deviation. Differences are HCM minus NOR with a 95% percentile-bootstrap confidence interval from 10 000 patient-level resamples. Positive Hedges' g indicates a larger value in HCM.

Endpoint	NOR	HCM	Difference (95% CI)	Hedges' g
Mean ED AHA-17 thickness (mm)	7.99 ± 0.88	11.66 ± 1.66	$+3.67$ [2.54, 4.67]	3.01
Maximum ED segment (mm)	9.07 ± 0.98	14.44 ± 1.94	$+5.37$ [4.11, 6.56]	3.82
Systolic thickening (percentage points)	53.2 ± 11.6	47.9 ± 11.5	-5.3 [-13.9, 3.0]	-0.45

The two end-diastolic confidence intervals exclude zero and their effect sizes are large, supporting descriptive separation in absolute thickness. In contrast, the interval for relative systolic thickening includes zero; this cohort does not establish a clear group difference in proportional thickening. Despite the maximum-segment contrast, none of the normal cases and only three of the ten HCM cases exceed the 15 mm clinical threshold. The measurements must therefore not be interpreted as a diagnostic result.

Figure 4.7 shows the patient-level spread behind these estimates. Absolute ED thickness separates the groups more clearly than relative systolic thickening, for which the distributions overlap.

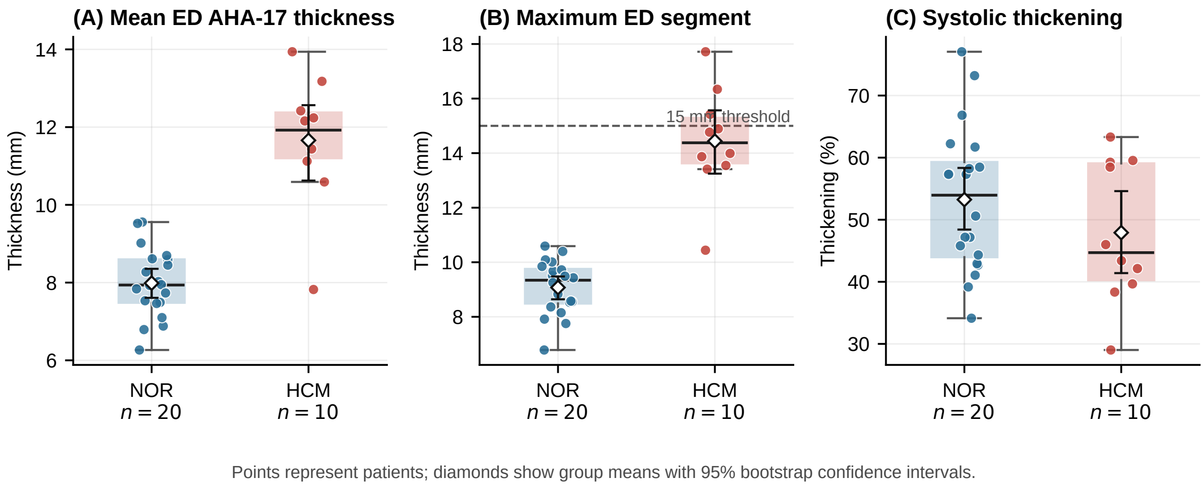

Figure 4.7. Patient-level AHA-17 wall-thickness outcomes for the normal and HCM groups. The dashed line marks the 15 mm threshold for the maximum ED segment; the analysis is descriptive and not a diagnostic validation.

One property of this analysis should be stated explicitly. Each patient contributes seventeen segment values, so the segment observations are not independent of one another. Segments from the same heart share the same reconstruction, the same input contours and the same anatomy. For this reason the segment values are used as a descriptive regional summary, and the group comparisons above are made at the patient level, where each patient counts once. The segment-level numbers should not be treated as if they came from independent observations.

4.4. Discussion

The results provide a direct answer to each research question stated in Section 1.3.

For **RQ1**, the model can reconstruct continuous, closed 3D LV surfaces from sparse SAX contours within the evaluated cohort. The surface errors are close to the in-plane resolution of the source images, and every extracted endocardial and epicardial surface is watertight after repair (Table 4.1). The continuous representation also removes much of the voxel staircasing visible in the segmentation-derived geometry (Figure 4.2). Compared with matched linear contour lofting, the model reduces the Chamfer distance, ASSD, and HD95 on both surfaces and improves myocardial overlap (Table 4.2). This provides evidence that the learned completion improves on direct interpolation between the observed contours. However, closure still depends on the repair stage described in Chapter 3; it is not guaranteed by surface extraction alone.

For **RQ2**, the model represents separate endocardial and epicardial surfaces at both ED and ES while maintaining a positive wall offset at every evaluated point. The paired surfaces do not collapse, and they reproduce the expected smaller cavity and thicker wall at ES. The answer is therefore positive at the level of geometric representation. Accuracy is not equal across phases: the model underestimates systolic thickening relative to the segmentation-derived geometry (Table 4.6). This is consistent with the use of ED-only synthetic pre-training and a binary phase condition.

For **RQ3**, measurements on the reconstructed surfaces track those from the derived geometry across patients, although their absolute values differ. The three non-ray methods show correlations of $r = 0.99$ between the two geometries and mean offsets of 0.58–0.91 mm, whereas the cone-ray method has a larger offset of 2.65 mm (Table 4.3). The reconstruction therefore preserves relative differences. At ED the absolute scale shifts upward for all four estimators; at ES the direction and magnitude of the shift depend on the method. This answers RQ3 in terms of computational agreement only, because the comparator is derived from the same segmentations and is not independent anatomical ground truth.

Taken together, the demonstrated result is geometric and computational: the model completes sparse contours into paired surfaces, preserves relative between-patient thickness variation, supports regional summaries, and performs better than a matched linear contour-lofting baseline. The observed ES shortfall may be explained by ED-only synthetic pre-training and binary phase conditioning, but their individual effects were not isolated by an ablation experiment. Likewise, the NOR–HCM separation in Table 4.8 shows biological plausibility, not diagnostic performance. The increasing cone-ray discrepancy at ES is consistent with the different measurement paths reviewed in Section 2.3.4; it does not establish that any estimator is anatomically correct.

4.5. Threats to Validity

The most important limitation concerns the reference. No public dataset provides independently acquired 3D ground-truth surfaces for these scans, so the reconstruction is evaluated against geometry *derived* from the segmentations. The Chamfer distances therefore measure agreement with a derived target rather than with true anatomy. The same caveat applies to the wall-thickness comparison: the segmentation is a stack of contours separated by several millimetres, so no source used in this chapter carries genuine through-plane information, and the way that gap is interpolated shifts the measured thickness. Absolute millimetre values should therefore not be over-interpreted; the comparisons that the data support are relative ones between pipelines evaluated on identical input. For the same reason, the wall-thickness study is an inter-method and reproducibility exercise on shared geometry, not a clinical validation. Without expert manual wall-thickness measurements, this work can claim an objective and locally resolved computational framework, but not the replacement of clinical measurement protocols.

Several further factors bound the external validity of the results, and the whole evaluation should be read as preliminary. The cohort holds 30 patients, 20 normal and 10 with hypertrophic cardiomyopathy, all reconstructed at end-diastole for the geometric metrics. This is enough to characterise the behaviour of the pipeline but far too small to support claims of clinical or pathological generalisation, and the hypertrophic subgroup in particular is only large enough for a directional observation. The regional analysis carries a further dependency: each patient contributes seventeen segments, so the segment values are repeated measures within a patient rather than independent samples, and they are reported as a descriptive summary for that reason. The relative thickening figures deserve particular caution, because dividing a systolic by a diastolic thickness amplifies small differences in where the wall is sampled. The regional summaries also depend on the AHA orientation, while all mesh-based measurements depend on the inference grid. A finer grid or an alternative anatomical frame could therefore shift the values. Only the decoder-offset map in Figure 4.4 is calibrated to the segmentation reference; the quantitative tables report geometric estimators before calibration. The reconstruction additionally inherits the quality of the input segmentation, and the training data span more than one collection, so some domain shift between acquisition sources is expected. Finally, the synthetic epicardium used during pre-training is a geometric construction rather than a measured surface, which the fine-tuning stage on real data is designed to correct but cannot fully rule out as a residual prior. A larger cohort, and ideally a dataset with genuine 3D ground truth, would be needed to confirm the results reported here.

The baseline comparison is deliberately narrow. Linear contour lofting tests whether the complete neural pipeline improves on direct between-slice interpolation under matched evaluation conditions. It does not cover spline lofting, shape-model fitting, Poisson reconstruction, or other learned models. No controlled ablation removes synthetic pre-training, local conditioning, phase conditioning, or positive surface coupling. The experiment therefore supports an advantage over one simple conventional baseline, but it cannot attribute that advantage to a specific component or establish that the full architecture is necessary.

CHAPTER 5

Conclusions

This thesis investigated the reconstruction of continuous three-dimensional left ventricular (LV) geometry from sparse short-axis (SAX) contours with an implicit neural representation. It also examined whether the reconstructed surfaces support computational estimates of myocardial wall thickness. The evaluation establishes what the proposed approach achieves and where the present evidence remains limited.

5.1. Answers to the Research Questions

For **RQ1**, the answer is positive within the evaluated cohort. The model reconstructs continuous endocardial and epicardial surfaces with mean errors close to the in-plane image resolution, and every evaluated surface is watertight after repair. The remaining systematic cavity underestimation shows that closed and visually smooth surfaces are not necessarily unbiased in volume.

For **RQ2**, the model represents both surfaces at ED and ES while maintaining a positive separation between their implicit fields. It reproduces the expected smaller cavity and thicker wall at ES. However, systolic thickening is lower than in the segmentation-derived comparator, so the result supports phase-conditioned representation more strongly than equal accuracy at both phases.

For **RQ3**, the reconstructed and segmentation-derived geometries preserve similar between-patient thickness variation for the field-based estimators, but their absolute values differ. The larger and phase-dependent disagreement of the cone-ray estimator also shows that wall thickness depends on the chosen measurement path. The answer is therefore one of computational agreement, not validation against true anatomy.

5.2. Contributions

The first contribution is a common contour-and-surface data pipeline for synthetic LV shapes and clinical ED/ES segmentations. The second is a phase-conditioned implicit model that combines global and local contour features and couples the epicardial field to the endocardial field through a positive offset. The third is an evaluation framework that connects surface and volume metrics with four local thickness estimators, continuous 3D maps, AHA-17 regional summaries, and patient-level NOR–HCM comparisons. Together, these components provide a reproducible way to study geometric completion and derived measurements from sparse SAX observations.

5.3. Limitations

As detailed in Section 4.5, the comparator is derived from the same sparse segmentations and is not independent 3D ground truth. Mesh repair contributes to the final watertight surfaces, and calibrated decoder-offset figures inherit their absolute scale from the segmentation reference. The cohort is small, especially for HCM, and the group analysis is descriptive. Finally, no matched comparison beyond linear contour lofting or controlled architectural ablation was completed. The current experiments show an improvement over direct linear interpolation, but they do not isolate the contribution of each model component or establish superiority over more advanced reconstruction methods.

5.4. Future Work

The first priority is evaluation against independent anatomical evidence, such as higher-resolution volumetric imaging or expert 3D surface and wall-thickness annotations, followed by validation on a larger external cohort. The second is a matched comparison with spline-based lofting and shape-model fitting, using the same patients, physical spacing, repair procedure, and metrics, together with point-wise error maps that localise failures at the apex, base, and highly curved wall regions. The third is a targeted experimental study of synthetic pre-training, local conditioning, positive surface coupling, and wall-offset supervision. Extending phase conditioning beyond a binary ED/ES label would then allow temporal consistency across the full cardiac cycle to be investigated.

The thesis therefore provides evidence that continuous paired LV geometry and locally resolved thickness summaries can be obtained from sparse SAX contours, while defining the additional comparisons and reference data required before clinical conclusions can be drawn.

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APPENDIX A

Supplementary Methodological Details

This appendix records supporting methodological material that would otherwise repeat information established in the main text.

A.1. Training data streams

All three training streams use the shared cache representation described in Section 3.2.5. Table A.1 provides a compact reference for their source, cardiac phase, and role in the two-stage training procedure, complementing the source datasets listed in Table 3.2.

Table A.1. Data streams sharing the cache schema and their role in training.

Stream	Phase	Source	Training role
Synthetic SSM	ED	UK Digital Heart Project LV SSM	Pre-training to establish a geometric shape prior
Real MRI	ED	ACDC and M&Ms-2	Fine-tuning on patient-specific diastolic anatomy
Real MRI	ES	ACDC and M&Ms-2	Fine-tuning on systolic anatomy with phase conditioning

Figure A.1 presents the same composition graphically.

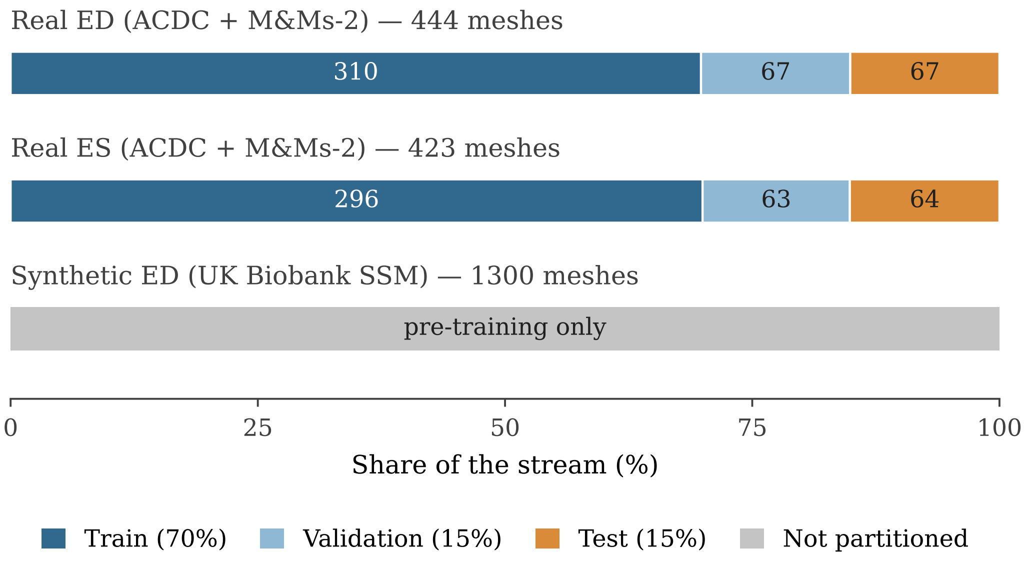

Figure A.1. Composition of the three data streams. Each bar shows how the valid samples of one stream are distributed over the 70 %/15 %/15 % train/validation/test partitions, with the number of patients annotated inside each segment. The synthetic stream, of which 1300 meshes were accepted out of 2048 sampled shape vectors, is reserved entirely for pre-training and is therefore not partitioned.

A.2. Real ED and ES mesh-preparation pipeline

Table A.2 lists, in order, the operations that convert a clinical segmentation stack into the watertight mesh used as derived supervision. The four stages of this conversion are described in Section 3.2.4.

Table A.2. Main operations in the real ED/ES mesh-preparation pipeline.

Operation	Purpose
RAS+ canonicalisation	Standardises patient-coordinate orientation while preserving physical spacing
Label harmonisation	Converts ACDC and M&Ms-2 to a common LV/myocardium mask
Isotropic resampling at 1.5 mm	Reduces anisotropy before 3D surface extraction
Per-slice cleaning	Removes small 2D fragments and fills holes
3D largest-component filtering	Keeps the main anatomical LV region
Gaussian mask smoothing and marching cubes	Suppresses staircase artefacts and extracts the surface
Taubin smoothing and Laplacian fairing	Reduces residual artefacts while preserving global shape
Basal cap perpendicular to the apex–base axis	Replaces the non-anatomical basal dome at the valve-plane boundary
Quadric decimation to 4000 vertices	Produces a consistently sized mesh for all cases
Plausibility checks (7 criteria)	Rejects geometrically implausible or corrupted cases